MIRA Control / 32 x 32 crossbar / Rev C engineering review

Existing host and board interface

INTERCONNECT

MCU

New acquisition, drive and processing circuitry

Zynq-7020 and memory

USB-C and FX20

DAC and ADC banks

BL and NL analog channels

Word line drivers

Interlocks and forming

Passive module interface

MIRA	
Sheet: /	
File: MIRA_Control.kicad_sch	
Title: MIRA Control | crossbar 32x32	
Size: A2	Date:
KiCad E.D.A. 10.0.8	Rev: C-review
	Id: 1/73

MCU — Host processor, memory and supervision

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

100_MCU_POWER_CLOCKS_AND_DEBUG

STM32 supply, LDO reservoir, clocks, reset and SWD; DONE/INIT status inputs.

200_MCU_FMC_AND_SDRAM

MCU external SDRAM and FMC signals, defaults and local bypass.

300_MCU_QSPI_MICROSD_AND_CONTROL

MCU QSPI flash, microSD card detect and supervisory GPIO/UART links.

Group scope

STM32 supply, clocks and debug; FMC SDRAM; QSPI, microSD and host control. The same physical MCU spans three leaves.

Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

MIRA

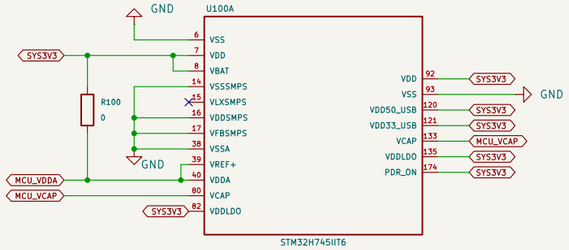
Sheet: /MCU/
File: MCU.kicad_sch

Title: Host processor, memory and supervision

Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 2/73

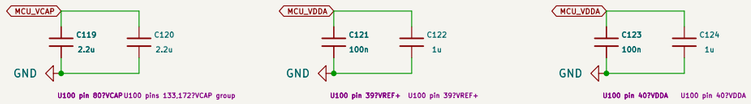
STM32 supply, clocks, reset and debug

Main MCU supply and core regulator | Rev.A

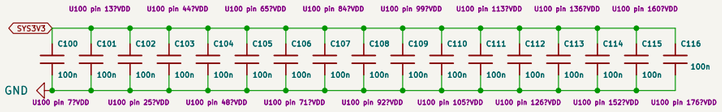


Main MCU supply bypass capacitors

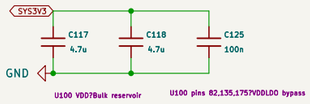
Core regulation and analog supply bypass



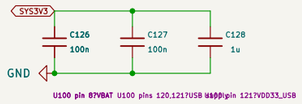
Main MCU supply bypass capacitors | Rev.A



VDD and LDO reservoir capacitors

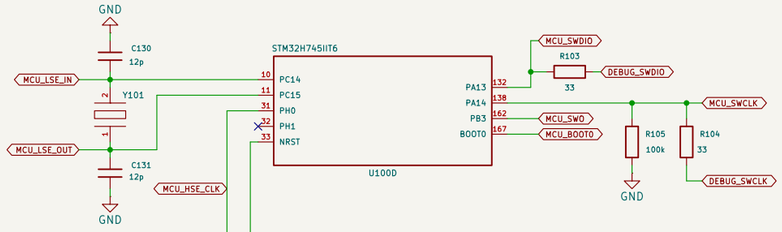


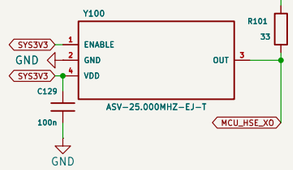
Backup and USB supply bypass



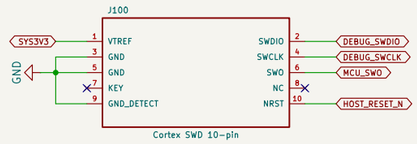
Main MCU clock references

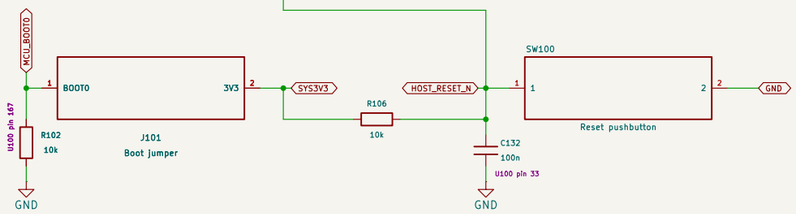
Main MCU clock references | Rev.A





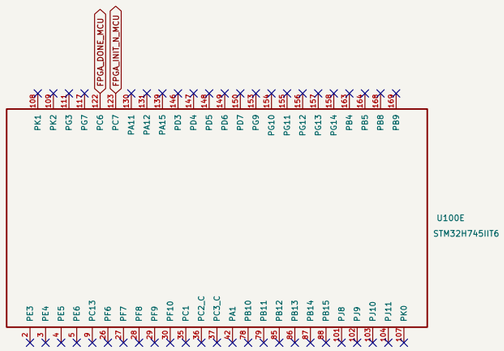
Main MCU reset, boot and SWD | Rev.A





MCU GPIO and FPGA configuration status

MCU GPIO and FPGA configuration status | Rev.A



Filtered supply: source declaration for ERC



Processor and supply domains

U100 is the STM32H745 host processor. SYS3V3 supplies VDD, VDDLDO, VBAT and the shown USB supply pins. R100 is a 0 ohm link feeding MCU_VDDA for VDDA and VREF+. The separate MCU units on the other sheets belong to this same physical package.

Internal core regulator

The three VCAP pins share MCU_VCAP, with C119 and C120 providing two 2.2 uF capacitors to ground. This net is the internal regulator reservoir, not an external supply output. VDDSMPS, VFBSMPS and VSSSMPS are grounded; VLXSMPS is left open in the drawn LDO supply configuration.

Local supply bypass

The 100 nF capacitors provide local bypass paths for the individual supply pins. The 4.7 uF capacitors support the shared rail. The analog, backup and USB supply groups have their own capacitors; their short return paths must be preserved during PCB placement.

Clock sources

Y100 provides a 25 MHz external clock through the 33 ohm R101 to MCU_HSE_CLK. Y101 is the 32.768 kHz LSE crystal, with 12 pF load capacitors C130 and C131. These two clock networks have different purposes and must be configured accordingly in firmware.

Reset, boot and access

R106 and C132 bias and filter HOST_RESET_N; SW100 pulls it low. R102 holds BOOT0 low, while J101 permits a high strap. J100 exposes SWD, SWO, reset and the supply reference. R103 and R104 are 33 ohm series resistors in SWD. PC6 (pin 122) receives FPGA_DONE_MCU and PC7 (pin 123) receives FPGA_INIT_N_MCU through the SYS3V3-powered U901 return buffer.

MIRA

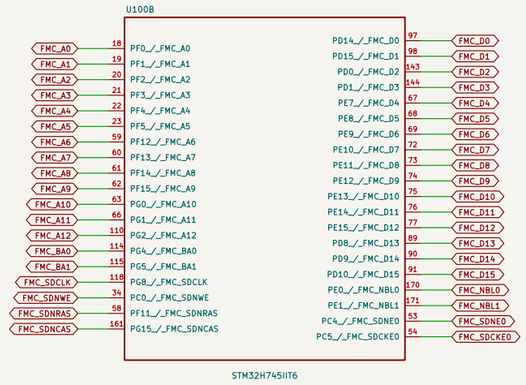
Sheet: /MCU/100_MCU_POWER_CLOCKS_AND_DEBUG/
File: 100_MCU_POWER_CLOCKS_AND_DEBUG.kicad_sch

Title: STM32 supply, clocks, reset and debug

Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 3/73

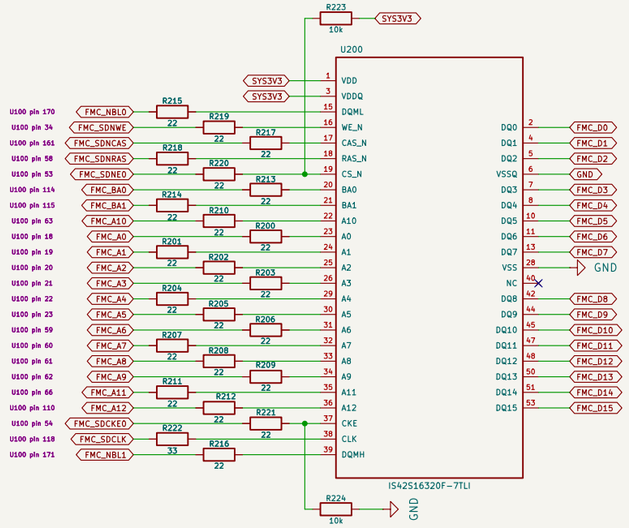
Main MCU FMC interface

Main MCU FMC interface | Rev.A

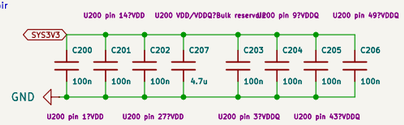


MAIN MCU SDRAM SOURCE DAMPING

Main MCU 64 MiB SDRAM | Rev.A



SDRAM supply bypass and reservoir



Memory path

The FMC unit of U100 drives U200, an IS42S16320F-7TLI SDRAM. The device provides 32M x 16 bits, or 64 MiB, of volatile storage. It belongs to the MCU address space and is separate from the FPGA DDR3L memory.

Bus organisation

FMC_D0 through FMC_D15 form the bidirectional data bus. FMC_A0 through FMC_A12 and FMC_BA0/FMC_BA1 select the row, column and internal bank according to the SDRAM command. FMC_NBL0 and FMC_NBL1 connect the two byte masks, allowing access to either half of the 16-bit word.

Commands and clock

FMC_SDNRAS, FMC_SDNCAS and FMC_SDNWE encode SDRAM commands. FMC_SDNE0 selects the device and FMC_SDCKE0 controls clock enable. FMC_SDCLK is routed through the separate 33 ohm R222. The other shown source-damping resistors are 22 ohms.

Startup bias and power

R223 pulls chip select high and R224 pulls clock enable low while the MCU pins are not actively driving them. U200 uses SYS3V3. Seven 100 nF capacitors and the 4.7 uF C207 provide local bypass and a supply reservoir.

Firmware and placement

Configure the FMC geometry, command sequence, timing parameters and refresh interval for this exact memory before application use. The schematic does not define those software values. Keep the clock, command and data routes controlled as a synchronous bus; retain the resistor-to-signal association shown here.

MIRA

Sheet: /MCU/200_MCU_FMC_AND_SDRAM/
File: 200_MCU_FMC_AND_SDRAM.kicad_sch

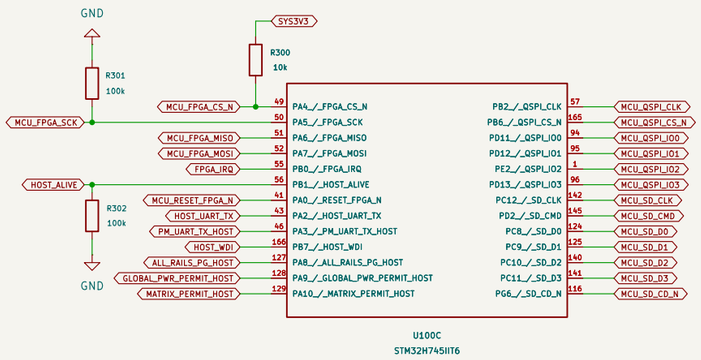
Title: STM32 FMC and 64 MiB SDRAM

Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 4/73

MCU nonvolatile storage and board control

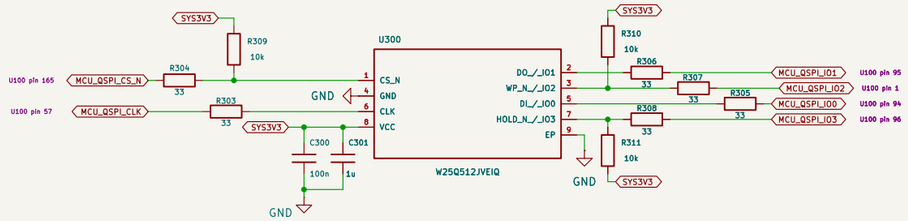
Main MCU serial memory and control

Main MCU serial memory and control | Rev.A



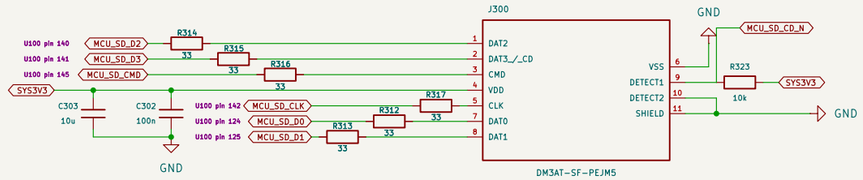
Main MCU 64 MiB QSPI NOR flash

Main MCU 64 MiB QSPI NOR flash | Rev.A

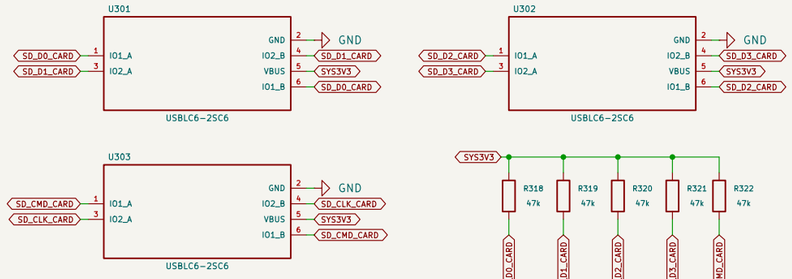


Main MCU microSD connector

Main MCU microSD connector | Rev.A



Main MCU microSD bias and ESD | Rev.A



QSPI program and data storage

U300 is a W25Q512JVEIQ NOR flash with 512 Mbit capacity, equivalent to 64 MiB. MCU_QSPI_CLK, MCU_QSPI_CS_N and four data signals connect it to U100. R303 through R308 are 33 ohm series resistors. The 10 kohm pull-ups keep chip select and the IO2/IO3 multifunction pins defined during startup.

Removable microSD storage

J300 connects a four-bit SD interface: clock, command and MCU_SD_D0 through MCU_SD_D3. The 33 ohm resistors separate MCU nets from SD_*_CARD nets. The 47 kohm resistors bias command and data. R323 pulls MCU_SD_CD_N high; the card-detect switch returns this net to ground. It reaches U100 PG6 (pin 116). C302 and C303 provide 100 nF and 10 uF local supply capacitance.

Connector protection

U301, U302 and U303 provide ESD paths for the six SD signal lines at the removable-card interface. The card shell and return contacts connect to the indicated ground network.

FPGA command interface

MCU_FPGA_SCK, MOSI, MISO and CS_N form the buffered command link, with FPGA_IRQ as an event return. MCU_RESET_FPGA_N reaches PROGRAM_B through the configuration-sheet open-drain buffer. FPGA_DONE_MCU and FPGA_INIT_N_MCU reach PC6/PC7 on the MCU supply sheet. Message formats and register maps must be implemented in firmware and FPGA logic.

Power-manager handshake

HOST_ALIVE is an application heartbeat: falling edges every 50 ms, maximum gap 100 ms. Stop pulses on lost link/session or task progress. Power sheet 5400 enforces a 170..230 ms timeout and first-edge readiness. HOST_WDI is separate MCU supervision. Neither heartbeat may use a free-running PWM/DMA. Status nets report the independent power manager.

MIRA

Sheet: /MCU/300_MCU_QSPI_MICROSD_AND_CONTROL/
File: 300_MCU_QSPI_MICROSD_AND_CONTROL.kicad_sch

Title: MCU nonvolatile storage and board control

Size: A3	Date: 2026-10-18	Rev: B
KiCad E.D.A. 10.0.8		Id: 5/73

INTERCONNECT — Power inputs and supervisory boundary

Consecutive circuit order; select a child sheet to inspect its circuit and sheet-specific explanation.

400_SYSTEM_POWER_AND_SUPERVISION_INTERFACE

Physical external rails, 3.3 V AON interface, Ioff status receivers and inhibit return.

410 DIGITAL POWER ENTRY

420 ANALOG POWER ENTRY

Group scope

Seven local power entry zones plus a separate supervision connector. SYS3V3 and IO3V3 receivers keep disabled receiver domains separate.

Reading guide

Each numeric child contains its own circuit explanation. Identical global labels connect between leaves. A shared U reference denotes one physical multi-unit device.

Physical references follow the first circuit leaf containing the component; multi-unit devices are not duplicated.

MIRA

Sheet: /INTERCONNECT/
File: INTERCONNECT.kicad_sch

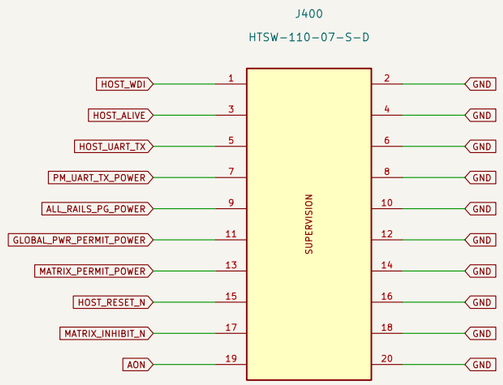
Title: Power inputs and supervisory boundary

Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 6/73

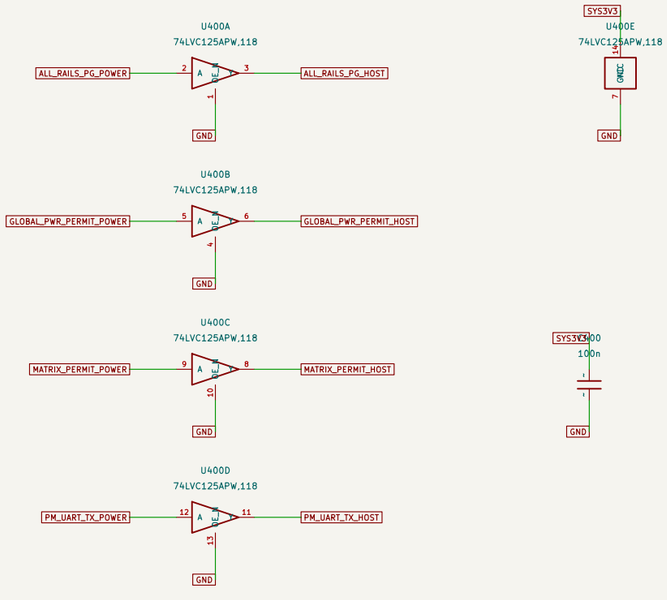
Power and supervision interface

Control-side pin assignment; AON = 3.3 V. Match the mating Power board before PCB release.

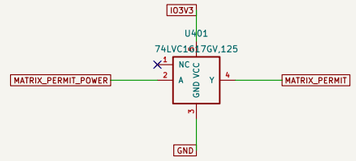
J400 – supervisor signals and AON



SYS3V3 receivers: powered-off isolation



IO3V3 permit receiver



Shared Power permit is received independently
in SYS3V3 and IO3V3; these output nets are not shorted.



Default states and open-drain supervision



Inputs from Power are 3.3 V logic. PG/permit default low; UART defaults high.
HOST_RESET_N is driven only by an open-drain reset source. MATRIX_INHIBIT_N is
an open-drain return to Power with an AON pull-up here. Watchdog/alive default low.

Distributed physical power inputs

J400: HTSW-110-07-S-D <-> Power J4901: SSW-110-01-S-D.
Power entry: J401..J407 on INTERCONNECT sheets 410 and 420.
Control headers B.Cu; Power sockets F.Cu; facing gap 12 mm.
Align each pair by pin 1 and pin 2 in common board coordinates.
No hot-plug; verify all eight pairs together before PCB release.

Separate receiving domains

U400 uses SYS3V3 for the MCU-side power-good, permit and UART signals. U401 uses IO3V3 for the matrix logic. Both Nexperia devices specify Ioff, so an AON-powered supervisor does not drive a disabled receiver domain through its outputs.

Permission and hardware inhibit

MATRIX_PERMIT_POWER is the common supervisor permission. Its two buffered outputs remain separate nets. The matrix receiver is pulled low on sheet 1300. MAT_FAULT_Q produces MATRIX_INHIBIT_N through Q2203; the new connector now exposes that output to the Power supervisor.

Required supply contract

AON is 3.3 V and starts before SYS3V3/IO3V3. VTT must source and sink at DDR1V35/2; DDR_VREF is a quiet reference at the same nominal level. P15/N15 and PROG_P15/PROG_N15 are separate analog domains. Signal mapping is defined here, not inferred from a missing mating board.

Power entry locations follow the consumers:
410: FPGA / DDR / MCU / USB.
420: DAC + matrix / ADC + analog front end.

Shared rails keep their existing names and source.
Place each local input beside its target decoupling.
Provide adjacent GND contacts and a continuous return plane.

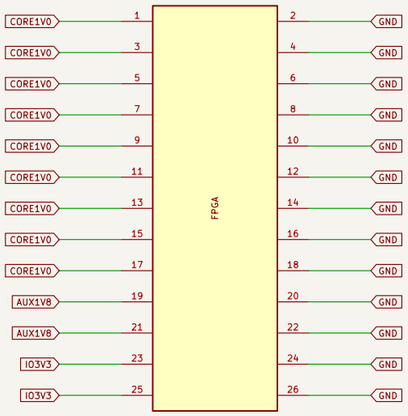
MIRA		
Sheet: /INTERCONNECT/400_SYSTEM_POWER_AND_SUPERVISION_INTERFACE/		
File: 400_SYSTEM_POWER_AND_SUPERVISION_INTERFACE.kicad_sch		
Title: Power and supervision interface		
Size: A3	Date: 2026-10-18	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 7/73

Distributed digital power: Power below / Control above

12 mm facing-surface gap. Sockets: Power F.Cu; headers: Control B.Cu. Place by consumer zone, not by schematic coordinates.

J401

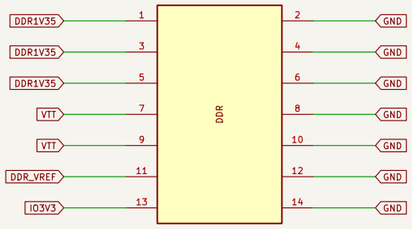
HTSW-113-07-S-D



J401 <-> J4902 | FPGA

J402

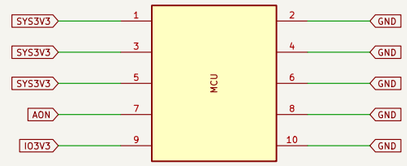
HTSW-107-07-S-D



J402 <-> J4903 | DDR

J403

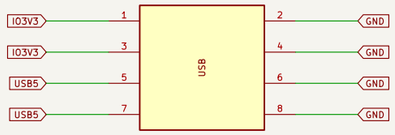
HTSW-105-07-S-D



J403 <-> J4904 | MCU

J404

HTSW-104-07-S-D



J404 <-> J4905 | USB

FPGA J4902 / J401:
Odd 1..15: eight CORE force contacts.
17: positive sense: route separately to FPGA.
19,21: AUX1V8. 23,25: IO3V3.
Every even pin: GND.

DDR VREF: keep quiet beside GND.
VTT must source and sink near termination.
Place Power converters below their zones.
Coordinate and height tolerances need checking.

MIRA

Sheet: /INTERCONNECT/410 DIGITAL POWER ENTRY/
File: 410_DISTRIBUTED_DIGITAL_POWER.kicad_sch

Title: MIRA Control – distributed digital power

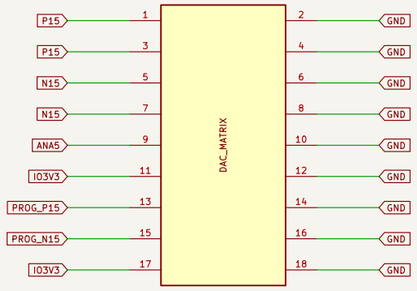
Size: A3	Date: 2026-10-28	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 31/73

Distributed analog power: Power below / Control above

12 mm facing-surface gap. Sockets: Power F.Cu; headers: Control B.Cu. Place by consumer zone, not by schematic coordinates.

J406

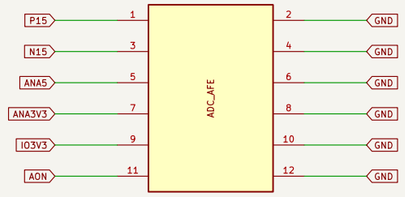
HTSW-109-07-S-D



J406 <-> J4907 | DAC_MATRIX

J407

HTSW-106-07-S-D



J407 <-> J4908 | ADC_AFE

DAC_MATRIX: place near DAC buffers and matrix power entry.
PROG_P15 / PROG_N15 remain switched supplies for the forming source.
ADC_AFE: place beside the TIA / ADC / reference power filters.

Repeated rail names are the same protected source, with multiple entry points.
Keep a continuous GND plane; local returns must not cross the FPGA/DDR load path.
Route quiet analog supply branches from the proper post-regulator/post-shunt output.
No connector-level isolation or independent zone protection is implied.

This sheet defines the electrical interface, not final XY placement or qualified current.
See docs/pcb-preparation/distributed-power for pinout, budgets and placement requirements.

MIRA

Sheet: /INTERCONNECT/420 ANALOG POWER ENTRY/
File: 420_DISTRIBUTED_ANALOG_POWER.kicad_sch

Title: MIRA Control — distributed analog power

Size: A3	Date: 2026-10-28	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 32/73

Zynq-7020 and memory

XC7Z020 | Bank 13, Bank 33, Bank 34, Bank 35

XC7Z020 | PS DDR 1, PS DDR 2, PS MIO 1, PS MIO 2

XC7Z020 | Configuration 1, Supply 1, Supply 2, Supply 3

PS DDR3L x16 | 512 MiB

PS clock, QSPI boot and host supervision

Zynq local supplies, analog PLL filter and I/O rail ingre

Sheet: /Zynq-7020 and memory/
File: C_GROUP_ZYNQ.kicad_sch

Title: Zynq-7020 and memory

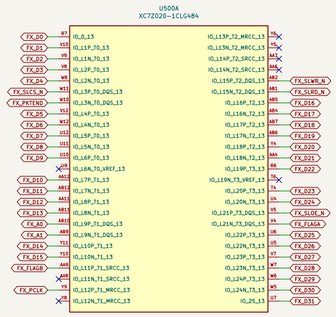
Size: A2 | Date: | **Rev: C-review**

KiCad E.D.A. 10.0.8 | Id: 60/73

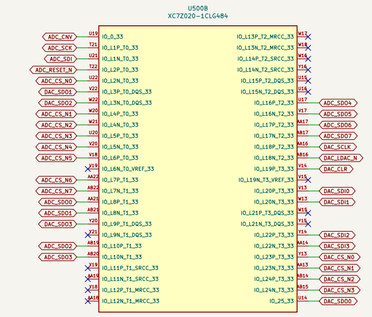
XC7Z020 | Bank 13, Bank 33, Bank 34, Bank 35

Physical ball numbers follow AMD XC7Z020 CLG484. Check Vivado I/O DRC, timing, power and startup before layout.

Bank 13



Bank 33



Bank 34

Bank 35

MIRA

Sheet: /Zynq-7020 and memory/XC7Z020 | Bank 13, Bank 33, Bank 34, Bank 35/
File: C_SOC0.kicad_sch

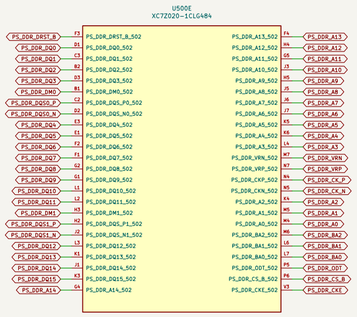
Title: XC7Z020 | Bank 13, Bank 33, Bank 34, Bank 35

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 61/73

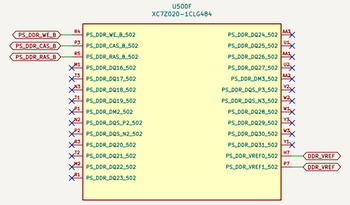
XC7Z020 | PS DDR 1, PS DDR 2, PS MIO 1, PS MIO 2

Physical ball numbers follow AMD XC7Z020 CLG484. Check Vivado I/O DRC, timing, power and startup before layout.

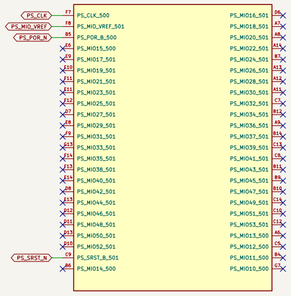
PS DDR 1



PS DDR 2



PS MIO 1



PS MIO 2



MIRA

Sheet: /Zynq-7020 and memory/XC7Z020 | PS DDR 1, PS DDR 2, PS MIO 1, PS MIO 2/
File: C_SOC1.kicad_sch

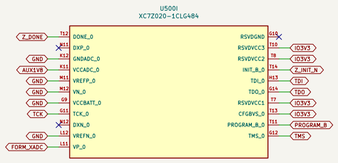
Title: XC7Z020 | PS DDR 1, PS DDR 2, PS MIO 1, PS MIO 2

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 62/73

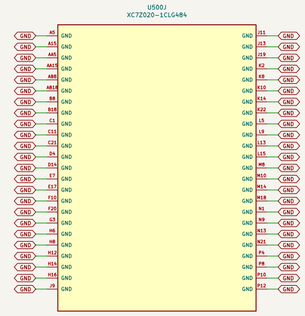
XC7Z020 | Configuration 1, Supply 1, Supply 2, Supply 3

Physical ball numbers follow AMD XC7Z020 CLG484. Check Vivado I/O DRC, timing, power and startup before layout.

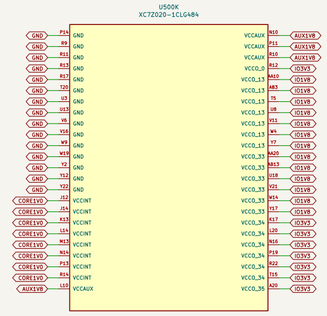
Configuration 1



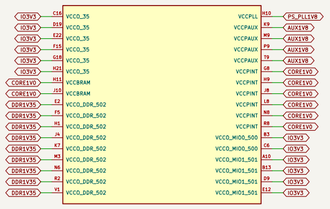
Supply 1



Supply 2



Supply 3



MIRA

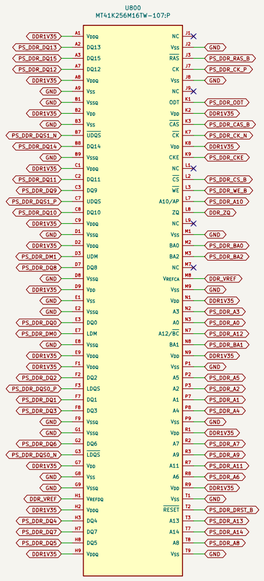
Sheet: /Zynq-7020 and memory/XC7Z020 | Configuration 1, Supply 1, Supply 2, Supply 3/
File: C_SOC2.kicad_sch

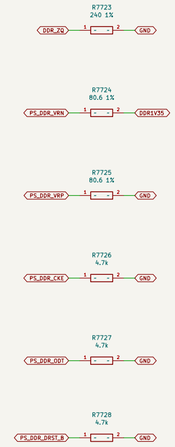
Title: XC7Z020 | Configuration 1, Supply 1, Supply 2, Supply 3

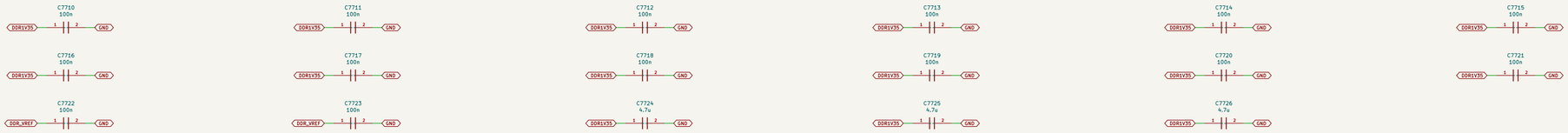
Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 63/73

PS DDR3L x16 | 512 MiB

Single rank. PS_DDR_CS_B on P6 drives CS# on L2. Impedance calibration uses 80 ohms for DDR3L.







MIRA

Sheet: /Zynq-7020 and memory/PS DDR3L x16 | 512 MiB/
File: C_DDR.kicad_sch

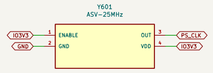
Title: PS DDR3L x16 | 512 MiB

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 64/73

PS clock, QSPI boot and host supervision

25 MHz single-ended PS clock. Separate boot flash from STM32H745. Both MIO banks at 3.3 V. QSPI straps sampled at PS_POR_B release.







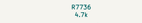


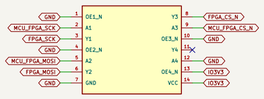


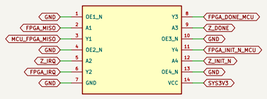








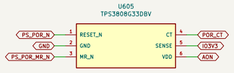




























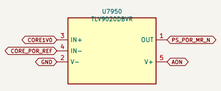




















MIRA

Sheet: /Zynq-7020 and memory/PS clock, QSPI boot and host supervision/
File: C_Boot.kicad_sch

Title: PS clock, QSPI boot and host supervision

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 65/73

Zynq local supplies, analog PLL filter and I/O rail ingress

Decoupling values are an initial allocation. Verify the local impedance profile with real mounting parasitics before layout.

L600
120 ohm at 100 MHz
AUX1V8 — PS_PLL1V8

C7733
10u
PS_PLL1V8 — GND

C7734
470n
PS_PLL1V8 — GND

C7735
100n
AUX1V8 — GND

#FLG0999
PWR_FEED
PS_PLL1V8 — POWER

CORE1V0

C7736 47u — CORE1V0 — GND
C7737 4.7u — CORE1V0 — GND
C7738 100n — CORE1V0 — GND
C7739 100n — CORE1V0 — GND
C7740 100n — CORE1V0 — GND
C7741 100n — CORE1V0 — GND
C7742 100n — CORE1V0 — GND
C7743 100n — CORE1V0 — GND
C7744 100n — CORE1V0 — GND
C7745 100n — CORE1V0 — GND
C7746 100n — CORE1V0 — GND
C7747 100n — CORE1V0 — GND

AUX1V8

C7748 47u — AUX1V8 — GND
C7749 4.7u — AUX1V8 — GND
C7750 100n — AUX1V8 — GND
C7751 100n — AUX1V8 — GND
C7752 100n — AUX1V8 — GND
C7753 100n — AUX1V8 — GND
C7754 100n — AUX1V8 — GND

IO1V8

C7755 47u — IO1V8 — GND
C7756 4.7u — IO1V8 — GND
C7757 100n — IO1V8 — GND
C7758 100n — IO1V8 — GND
C7759 100n — IO1V8 — GND
C7760 100n — IO1V8 — GND
C7761 100n — IO1V8 — GND

IO3V3

C7762 47u — IO3V3 — GND
C7763 4.7u — IO3V3 — GND
C7764 100n — IO3V3 — GND
C7765 100n — IO3V3 — GND
C7766 100n — IO3V3 — GND
C7767 100n — IO3V3 — GND
C7768 100n — IO3V3 — GND

DDR1V35

C7769 47u — DDR1V35 — GND
C7770 4.7u — DDR1V35 — GND
C7771 100n — DDR1V35 — GND
C7772 100n — DDR1V35 — GND
C7773 100n — DDR1V35 — GND
C7774 100n — DDR1V35 — GND
C7775 100n — DDR1V35 — GND

J405
HTSW-105-07-S-D
1 IO1V8 — 2 GND — 3 IO1V8 — 4 GND — 5 IO3V3
6 GND — 7 IO3V3 — 8 GND — 9 IO3V3 — 10 GND

TP610
ANA3V3 test
ANA3V3 — TP

TP611
PROG_P15 test
PROG_P15 — TP

TP612
PROG_N15 test
PROG_N15 — TP

MIRA
Sheet: /Zynq—7020 and memory/Zynq local supplies, analog PLL filter and I/O rail ingre/
File: C_Decoupling.kicad_sch
Title: Zynq local supplies, analog PLL filter and I/O rail ingress
Size: A2 | Date: | Rev: C—review
KiCad E.D.A. 10.0.8 | Id: 66/73

USB-C and FX20

FX20 | parallel interface, USB PHY and supplies

USB-C | two SuperSpeed lanes, USB 2.0 and attach detectio

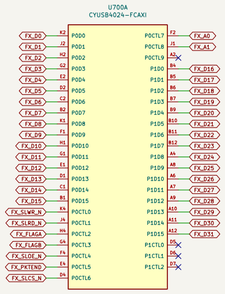
Sheet: /USB-C and FX20/
File: C_GROUP_USB.kicad_sch

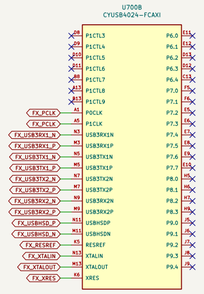
Title: USB-C and FX20

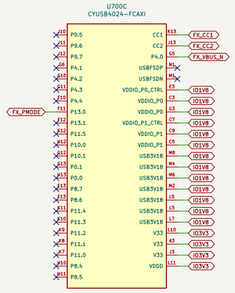
Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 67/73

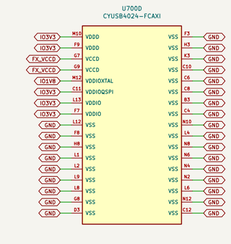
FX20 | parallel interface, USB PHY and supplies

SDR 32-bit wide link uses PCLK. Control pin assignment is conditional on the GPIF state machine and bidirectional turnaround timing.



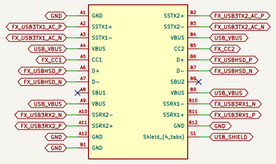


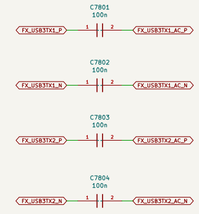


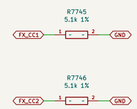

U700D
CYUSB4024-FCAXI
IO3V3
FX_VCCD
FX_VCCD
IO1V8
IO3V3
IO3V3
GND
GND
GND
GND
GND
GND
GND
GND
VDDD
VDDD
VDDIOXTAL
VDDIODQSP1
VDDIO
VSS
VSS
VSS

USB-C | two SuperSpeed lanes, USB 2.0 and attach detection

Self-powered upstream port. Two Rd terminations for orientation, AC coupling on TX only, no external SuperSpeed lane switch.













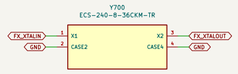












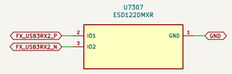














Review crystal load against selected crystal and stray capacitance. Verify connector loss, ESD package capacitance, ground strategy and Gen2x2 compliance.

































































MIRA

Sheet: /USB-C and FX20/USB-C | two SuperSpeed lanes, USB 2.0 and attach detectio/
File: C_USB_C.kicad_sch

Title: USB-C | two SuperSpeed lanes, USB 2.0 and attach detection

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 69/73

DAC and ADC banks

BL0..BL15 | stimulus and paired sampling

BL16..BL31 | stimulus and paired sampling

NL0..NL15 | stimulus and paired sampling

NL16..NL31 | stimulus and paired sampling

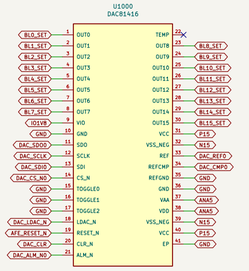
Sheet: /DAC and ADC banks/
File: C_GROUP_CONVERTERS.kicad_sch

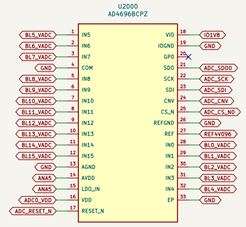
Title: DAC and ADC banks

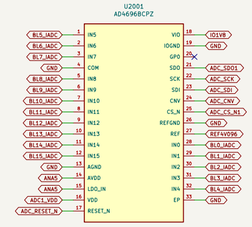
Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 70/73

BL0..BL15 | stimulus and paired sampling

16 independent DAC outputs. Voltage and current ADCs select the same index on a shared CNV edge.

















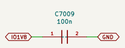




























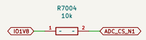












VIO = 1.8 V throughout bank 33. Local REF reservoir per ADC. Use enhanced high-Z / acquisition timing only after characterization.







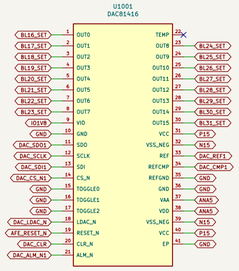


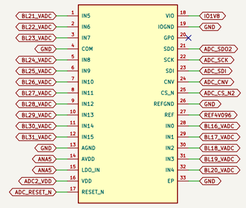


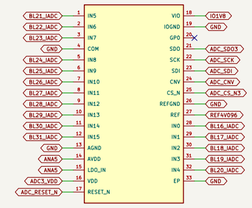
MIRA
Sheet: /DAC and ADC banks/BL0..BL15 | stimulus and paired sampling/
File: C_Bank0.kicad_sch
Title: BL0..BL15 | stimulus and paired sampling
Size: A2
Date:
Rev: C-review
KiCad E.D.A. 10.0.8
Id: 71/73

BL16..BL31 | stimulus and paired sampling

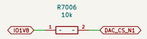
16 independent DAC outputs. Voltage and current ADCs select the same index on a shared CNV edge.













































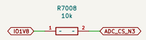












VIO = 1.8 V throughout bank 33. Local REF reservoir per ADC. Use enhanced high-Z / acquisition timing only after characterization.

MIRA

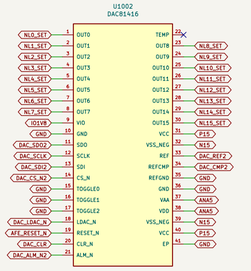
Sheet: /DAC and ADC banks/BL16..BL31 | stimulus and paired sampling/
File: C_Bank1.kicad_sch

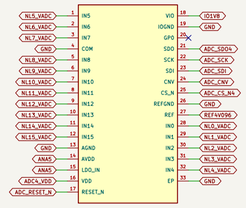
Title: BL16..BL31 | stimulus and paired sampling

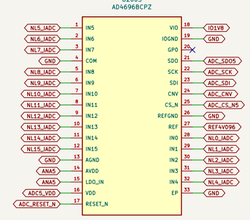
Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 72/73

NL0..NL15 | stimulus and paired sampling

16 independent DAC outputs. Voltage and current ADCs select the same index on a shared CNV edge.

























































VIO = 1.8 V throughout bank 33. Local REF reservoir per ADC. Use enhanced high-Z / acquisition timing only after characterization.

MIRA

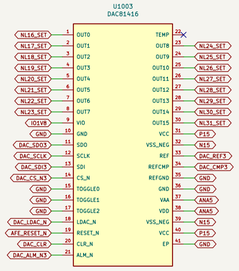
Sheet: /DAC and ADC banks/NL0..NL15 | stimulus and paired sampling/
File: C_Bank2.kicad_sch

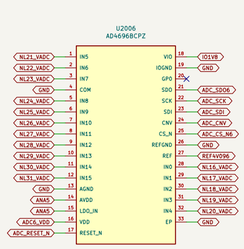
Title: NL0..NL15 | stimulus and paired sampling

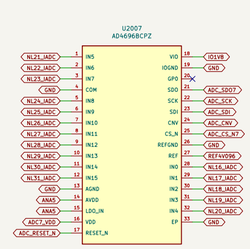
Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 73/73

NL16..NL31 | stimulus and paired sampling

16 independent DAC outputs. Voltage and current ADCs select the same index on a shared CNV edge.

























































VIO = 1.8 V throughout bank 33. Local REF reservoir per ADC. Use enhanced high-Z / acquisition timing only after characterization.

MIRA

Sheet: /DAC and ADC banks/NL16..NL31 | stimulus and paired sampling/
File: C_Bank3.kicad_sch

Title: NL16..NL31 | stimulus and paired sampling

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 74/73

BL and NL analog channels

BL0 .. BL3 | drive, remote sense and signed current

BL4 .. BL7 | drive, remote sense and signed current

BL8 .. BL11 | drive, remote sense and signed current

BL12 .. BL15 | drive, remote sense and signed current

BL16 .. BL19 | drive, remote sense and signed current

BL20 .. BL23 | drive, remote sense and signed current

BL24 .. BL27 | drive, remote sense and signed current

BL28 .. BL31 | drive, remote sense and signed current

NL0 .. NL3 | drive, remote sense and signed current

NL4 .. NL7 | drive, remote sense and signed current

NL8 .. NL11 | drive, remote sense and signed current

NL12 .. NL15 | drive, remote sense and signed current

NL16 .. NL19 | drive, remote sense and signed current

NL20 .. NL23 | drive, remote sense and signed current

NL24 .. NL27 | drive, remote sense and signed current

NL28 .. NL31 | drive, remote sense and signed current

How to read a BL or NL channel

Sheet: /BL and NL analog channels/
File: C_GROUP_LINES.kicad_sch

Title: BL and NL analog channels

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 75/73

BL0 .. BL3 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

U7001
OPA4192
C7109
100n
C7110
100n
BL0
U7002
OPA4192
C7111
100n
C7112
100n
BL1
U7003
ADG442
C7113
100n
C7114
100n
BL2
U7004
TMUX6112
C7115
100n
C7116
100n
BL3
R7017
5.00 0.1%
R7018
1.00k
R7019
100k
R7020
20.0k
R7021
10.0k
R7022
47
R7023
47k
C7085
220p C0G
C7086
220p C0G
C7087
1n
R7025
5.00 0.1%
R7026
1.00k
R7027
100k
R7028
20.0k
R7029
10.0k
R7030
47
R7031
47k
C7091
220p C0G
C7092
220p C0G
C7093
1n
R7033
5.00 0.1%
R7034
1.00k
R7035
100k
R7036
20.0k
R7037
10.0k
R7038
47
R7039
47k
C7097
220p C0G
C7098
220p C0G
C7099
1n
R7041
5.00 0.1%
R7042
1.00k
R7043
100k
R7044
20.0k
R7045
10.0k
R7046
47
R7047
47k
C7103
220p C0G
C7104
220p C0G
C7105
1n
U7005
INA296A2
U7008
INA296A2
U7011
INA296A2
U7014
INA296A2
U7006
TLV3202
U7009
TLV3202
U7012
TLV3202
U7015
TLV3202
D7001
BAT54A
D7002
BAT54A
D7003
BAT54A
D7004
BAT54A
R7024
4.7k
R7032
4.7k
R7040
4.7k
R7048
4.7k
U7007
SN74LVC1G74
U7010
SN74LVC1G74
U7013
SN74LVC1G74
U7016
SN74LVC1G74
C7088
100n
C7089
100n
C7090
100n
C7094
100n
C7095
100n
C7096
100n
C7100
100n
C7101
100n
C7102
100n
C7106
100n
C7107
100n
C7108
100n

MIRA
Sheet: /BL and NL analog channels/BL0 .. BL3 | drive, remote sense and signed current/
File: C_AFE_BL0_BL3.kicad_sch
Title: BL0 .. BL3 | drive, remote sense and signed current
Size: A2
Date:
Rev: C-review
KiCad E.D.A. 10.0.8
Id: 76/73

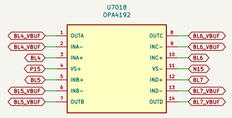
BL4 .. BL7 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin. IADC measures current through the 5 ohm shunt.



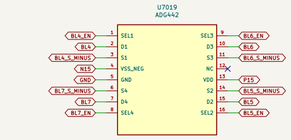






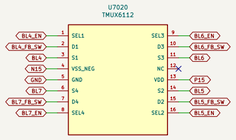








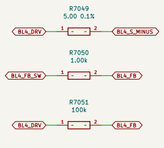


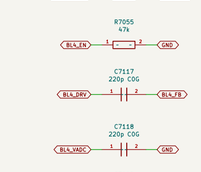


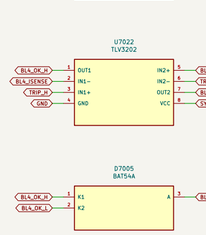




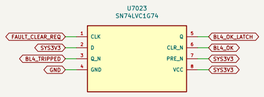
BL4





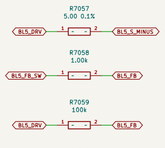


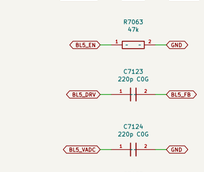


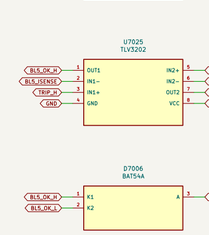




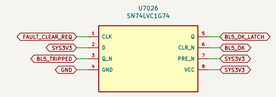
BL5





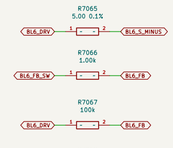


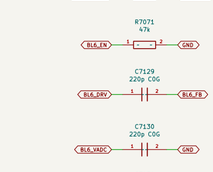


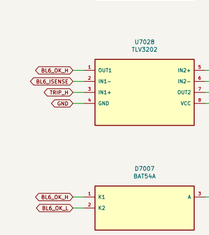




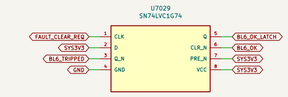
BL6





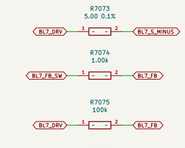


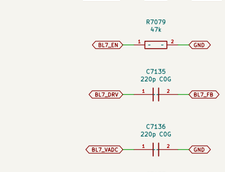


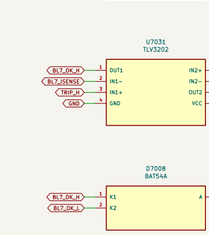




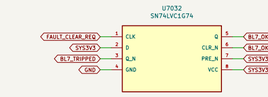
BL7













MIRA

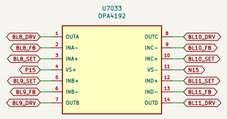
Sheet: /BL and NL analog channels/BL4 .. BL7 | drive, remote sense and signed current/
File: C_AFE_BL4_BL7.kicad_sch

Title: BL4 .. BL7 | drive, remote sense and signed current

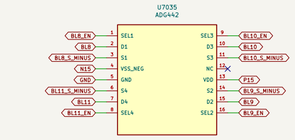
Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 77/73

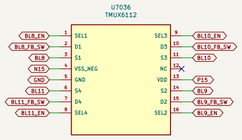
BL8 .. BL11 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin. IADC measures current through the 5 ohm shunt.























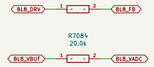


BL8











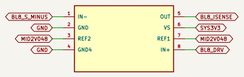








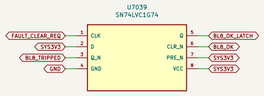












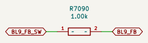




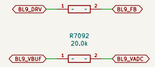


BL9



















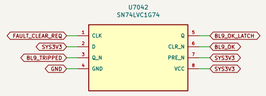












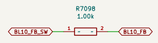




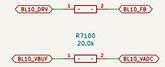


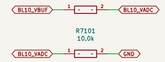
BL10











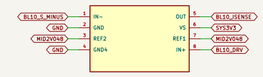


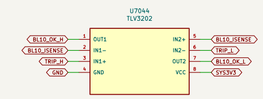




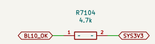


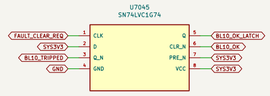
















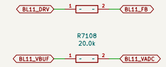


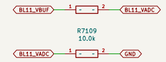
BL11

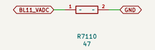










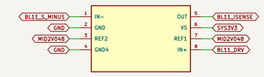








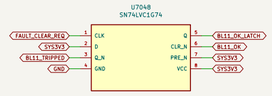


















MIRA

Sheet: /BL and NL analog channels/BL8 .. BL11 | drive, remote sense and signed current/
File: C_AFE_BL8_BL11.kicad_sch

Title: BL8 .. BL11 | drive, remote sense and signed current

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 78/73

BL12 .. BL15 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

U7049
OPA4192
BL12_DRV
BL12_FB
BL12_SET
P15
BL13_SET
BL13_FB
BL13_DRV
OUTA
INA−
INA+
VS+
INB+
INB−
OUTB
OUTC
INC−
INC+
VS−
IND+
IND−
OUTD
BL14_DRV
BL14_FB
BL14_SET
N15
BL15_SET
BL15_FB
BL15_DRV
C7205
100n
P15
GND
C7206
100n
N15
GND
U7050
OPA4192
BL12_VBUF
BL12_VBUF
BL12
P15
BL13
BL13_VBUF
BL13_VBUF
OUTA
INA−
INA+
VS+
INB+
INB−
OUTB
OUTC
INC−
INC+
VS−
IND+
IND−
OUTD
BL14_VBUF
BL14_VBUF
BL14
N15
BL15
BL15_VBUF
BL15_VBUF
C7207
100n
P15
GND
C7208
100n
N15
GND
U7051
ADG442
BL12_EN
BL12
BL12_S_MINUS
N15
GND
BL15_S_MINUS
BL15
BL15_EN
S1
VSS_NEG
GND
S4
D4
SEL4
SEL1
D1
S3
NC
VDD
S2
D2
SEL2
SEL3
D3
BL14_EN
BL14
BL14_S_MINUS
P15
BL13_S_MINUS
BL13
BL13_EN
C7209
100n
P15
GND
C7210
100n
N15
GND
U7052
TMUX6112
BL12_EN
BL12_FB_SW
BL12
N15
GND
BL15
BL15_FB_SW
BL15_EN
S1
VSS_NEG
GND
S4
D4
SEL4
SEL1
D1
S3
NC
VDD
S2
D2
SEL2
SEL3
D3
BL14_EN
BL15_FB_SW
BL14
P15
BL13
BL13_FB_SW
BL13_EN
C7211
100n
P15
GND
C7212
100n
N15
GND

BL12

R7113
5.00 0.1%
BL12_DRV
BL12_S_MINUS
R7114
1.00k
BL12_FB_SW
BL12_FB
R7115
100k
BL12_DRV
BL12_FB
R7116
20.0k
BL12_VBUF
BL12_VADC
R7117
10.0k
BL12_VADC
GND
R7118
47
BL12_ISENSE
BL12_IADC
R7119
47k
BL12_EN
GND
C7181
220p C0G
BL12_DRV
BL12_FB
C7182
220p C0G
BL12_VADC
GND
C7183
1n
BL12_IADC
GND

BL13

R7121
5.00 0.1%
BL13_DRV
BL13_S_MINUS
R7122
1.00k
BL13_FB_SW
BL13_FB
R7123
100k
BL13_DRV
BL13_FB
R7124
20.0k
BL13_VBUF
BL13_VADC
R7125
10.0k
BL13_VADC
GND
R7126
47
BL13_ISENSE
BL13_IADC
R7127
47k
BL13_EN
GND
C7187
220p C0G
BL13_DRV
BL13_FB
C7188
220p C0G
BL13_VADC
GND
C7189
1n
BL13_IADC
GND

BL14

R7129
5.00 0.1%
BL14_DRV
BL14_S_MINUS
R7130
1.00k
BL14_FB_SW
BL14_FB
R7131
100k
BL14_DRV
BL14_FB
R7132
20.0k
BL14_VBUF
BL14_VADC
R7133
10.0k
BL14_VADC
GND
R7134
47
BL14_ISENSE
BL14_IADC
R7135
47k
BL14_EN
GND
C7193
220p C0G
BL14_DRV
BL14_FB
C7194
220p C0G
BL14_VADC
GND
C7195
1n
BL14_IADC
GND

BL15

R7137
5.00 0.1%
BL15_DRV
BL15_S_MINUS
R7138
1.00k
BL15_FB_SW
BL15_FB
R7139
100k
BL15_DRV
BL15_FB
R7140
20.0k
BL15_VBUF
BL15_VADC
R7141
10.0k
BL15_VADC
GND
R7142
47
BL15_ISENSE
BL15_IADC
R7143
47k
BL15_EN
GND
C7199
220p C0G
BL15_DRV
BL15_FB
C7200
220p C0G
BL15_VADC
GND
C7201
1n
BL15_IADC
GND

U7053
INA296A2
BL12_S_MINUS
GND
MID2V048
GND
IN−
GND
REF2
GND4
OUT
VS
REF1
IN+
BL12_ISENSE
SYS3V3
MID2V048
BL12_DRV
U7056
INA296A2
BL13_S_MINUS
GND
MID2V048
GND
IN−
GND
REF2
GND4
OUT
VS
REF1
IN+
BL13_ISENSE
SYS3V3
MID2V048
BL13_DRV
U7059
INA296A2
BL14_S_MINUS
GND
MID2V048
GND
IN−
GND
REF2
GND4
OUT
VS
REF1
IN+
BL14_ISENSE
SYS3V3
MID2V048
BL14_DRV
U7062
INA296A2
BL15_S_MINUS
GND
MID2V048
GND
IN−
GND
REF2
GND4
OUT
VS
REF1
IN+
BL15_ISENSE
SYS3V3
MID2V048
BL15_DRV

U7054
TLV3202
BL12_OK_H
BL12_ISENSE
TRIP_H
GND
OUT1
IN1−
IN1+
GND
IN2+
IN2−
OUT2
VCC
BL12_ISENSE
TRIP_L
BL12_OK_L
SYS3V3
U7057
TLV3202
BL13_OK_H
BL13_ISENSE
TRIP_H
GND
OUT1
IN1−
IN1+
GND
IN2+
IN2−
OUT2
VCC
BL13_ISENSE
TRIP_L
BL13_OK_L
SYS3V3
U7060
TLV3202
BL14_OK_H
BL14_ISENSE
TRIP_H
GND
OUT1
IN1−
IN1+
GND
IN2+
IN2−
OUT2
VCC
BL14_ISENSE
TRIP_L
BL14_OK_L
SYS3V3
U7063
TLV3202
BL15_OK_H
BL15_ISENSE
TRIP_H
GND
OUT1
IN1−
IN1+
GND
IN2+
IN2−
OUT2
VCC
BL15_ISENSE
TRIP_L
BL15_OK_L
SYS3V3

D7013
BAT54A
BL12_OK_H
BL12_OK_L
K1
K2
A
BL12_OK
D7014
BAT54A
BL13_OK_H
BL13_OK_L
K1
K2
A
BL13_OK
D7015
BAT54A
BL14_OK_H
BL14_OK_L
K1
K2
A
BL14_OK
D7016
BAT54A
BL15_OK_H
BL15_OK_L
K1
K2
A
BL15_OK
R7120
4.7k
BL12_OK
SYS3V3
R7128
4.7k
BL13_OK
SYS3V3
R7136
4.7k
BL14_OK
SYS3V3
R7144
4.7k
BL15_OK
SYS3V3

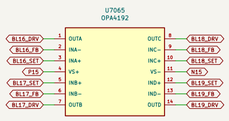
U7055
SN74LVC1G74
FAULT_CLEAR_REQ
SYS3V3
BL12_TRIPPED
GND
CLK
D
Q_N
GND
Q
CLR_N
PRE_N
VCC
BL12_OK_LATCH
BL12_OK
SYS3V3
SYS3V3
U7058
SN74LVC1G74
FAULT_CLEAR_REQ
SYS3V3
BL13_TRIPPED
GND
CLK
D
Q_N
GND
Q
CLR_N
PRE_N
VCC
BL13_OK_LATCH
BL13_OK
SYS3V3
SYS3V3
U7061
SN74LVC1G74
FAULT_CLEAR_REQ
SYS3V3
BL14_TRIPPED
GND
CLK
D
Q_N
GND
Q
CLR_N
PRE_N
VCC
BL14_OK_LATCH
BL14_OK
SYS3V3
SYS3V3
U7064
SN74LVC1G74
FAULT_CLEAR_REQ
SYS3V3
BL15_TRIPPED
GND
CLK
D
Q_N
GND
Q
CLR_N
PRE_N
VCC
BL15_OK_LATCH
BL15_OK
SYS3V3
SYS3V3

C7184
100n
SYS3V3
GND
C7185
100n
SYS3V3
GND
C7186
100n
SYS3V3
GND
C7190
100n
SYS3V3
GND
C7191
100n
SYS3V3
GND
C7192
100n
SYS3V3
GND
C7196
100n
SYS3V3
GND
C7197
100n
SYS3V3
GND
C7198
100n
SYS3V3
GND
C7202
100n
SYS3V3
GND
C7203
100n
SYS3V3
GND
C7204
100n
SYS3V3
GND

MIRA
Sheet: /BL and NL analog channels/BL12 .. BL15 | drive, remote sense and signed current/
File: C_AFE_BL12_BL15.kicad_sch
Title: BL12 .. BL15 | drive, remote sense and signed current
Size: A2
Date:
Rev: C−review
KiCad E.D.A. 10.0.8
Id: 79/73

BL16 .. BL19 | drive, remote sense and signed current

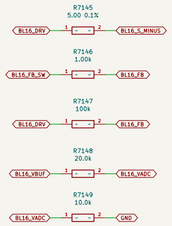
Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

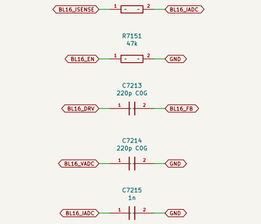


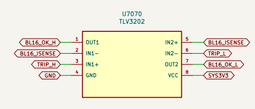




BL16







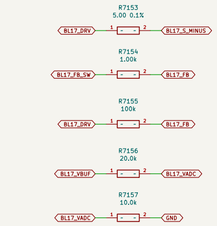


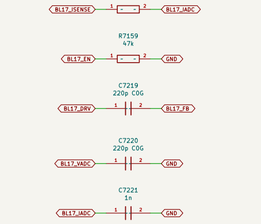


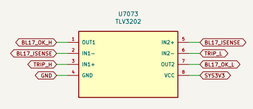




BL17

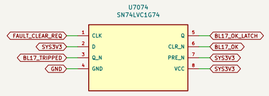






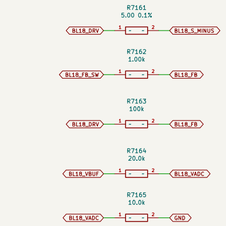


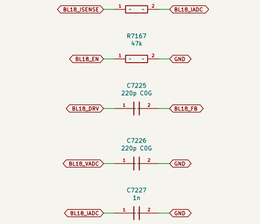


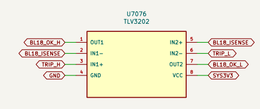




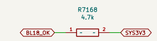
BL18

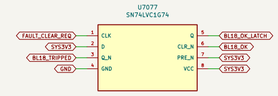






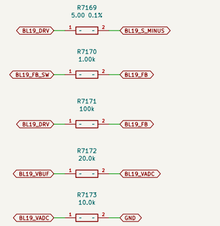


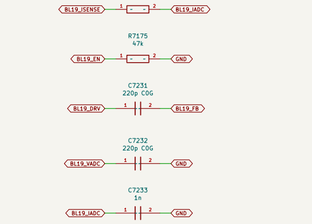


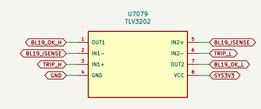




BL19

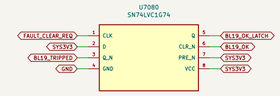














MIRA
Sheet: /BL and NL analog channels/BL16 .. BL19 | drive, remote sense and signed current/
File: C_AFE_BL16_BL19.kicad_sch

Title: BL16 .. BL19 | drive, remote sense and signed current

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 80/73

BL20 .. BL23 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

BL20

U7081
OPA4192
C7269
100n
C7270
100n
R7177
5.00 0.1%
R7178
1.00k
R7179
100k
R7180
20.0k
R7181
10.0k
R7182
47
R7183
47k
C7245
220p C0G
C7246
220p C0G
C7247
1n
U7085
INA296A2
U7086
TLV3202
D7021
BAT54A
R7184
4.7k
U7087
SN74LVC1G74
C7248
100n
C7249
100n
C7250
100n

BL21

U7082
OPA4192
C7271
100n
C7272
100n
R7185
5.00 0.1%
R7186
1.00k
R7187
100k
R7188
20.0k
R7189
10.0k
R7190
47
R7191
47k
C7251
220p C0G
C7252
220p C0G
C7253
1n
U7088
INA296A2
U7089
TLV3202
D7022
BAT54A
R7192
4.7k
U7090
SN74LVC1G74
C7254
100n
C7255
100n
C7256
100n

BL22

U7083
AD6442
C7273
100n
C7274
100n
R7193
5.00 0.1%
R7194
1.00k
R7195
100k
R7196
20.0k
R7197
10.0k
R7198
47
R7199
47k
C7257
220p C0G
C7258
220p C0G
C7259
1n
U7091
INA296A2
U7092
TLV3202
D7023
BAT54A
R7200
4.7k
U7093
SN74LVC1G74
C7260
100n
C7261
100n
C7262
100n

BL23

U7084
TMUX6112
C7275
100n
C7276
100n
R7201
5.00 0.1%
R7202
1.00k
R7203
100k
R7204
20.0k
R7205
10.0k
R7206
47
R7207
47k
C7263
220p C0G
C7264
220p C0G
C7265
1n
U7094
INA296A2
U7095
TLV3202
D7024
BAT54A
R7208
4.7k
U7096
SN74LVC1G74
C7266
100n
C7267
100n
C7268
100n

MIRA
Sheet: /BL and NL analog channels/BL20 .. BL23 | drive, remote sense and signed current/
File: C_AFE_BL20_BL23.kicad_sch
Title: BL20 .. BL23 | drive, remote sense and signed current
Size: A2
Date:
Rev: C-review
KiCad E.D.A. 10.0.8
Id: 81/73

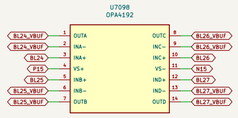
BL24 .. BL27 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.



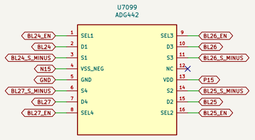






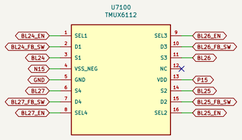










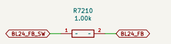




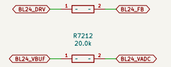


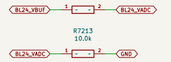
BL24













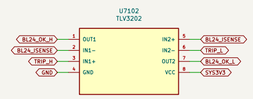




















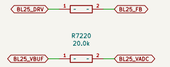


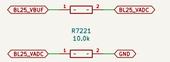
BL25











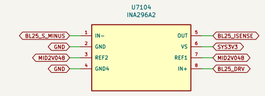


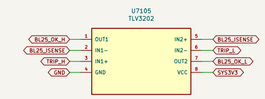




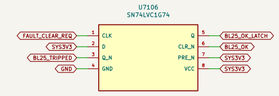












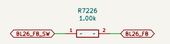




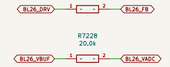


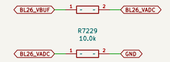
BL26











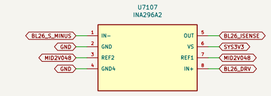


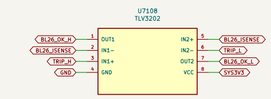


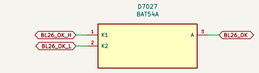


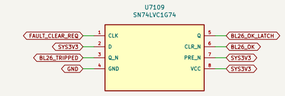












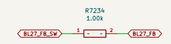


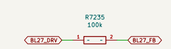


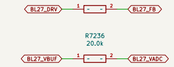


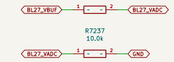
BL27

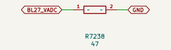










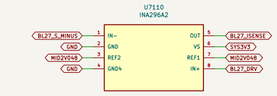


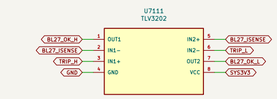


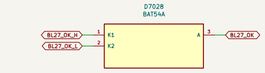


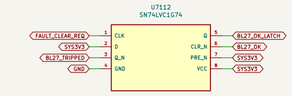














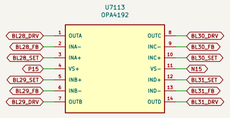




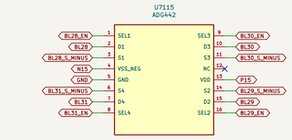
MIRA
Sheet: /BL and NL analog channels/BL24 .. BL27 | drive, remote sense and signed current/
File: C_AFE_BL24_BL27.kicad_sch
Title: BL24 .. BL27 | drive, remote sense and signed current
Size: A2 | Date: | Rev: C-review
KiCad E.D.A. 10.0.8 | Id: 82/73

BL28 .. BL31 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

























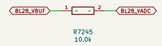
BL28











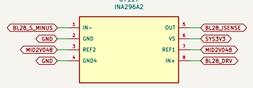


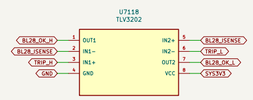






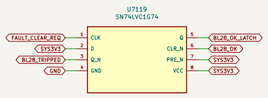


















BL29





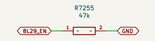






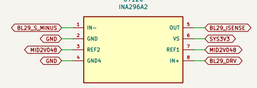


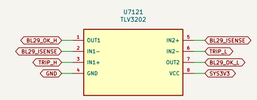




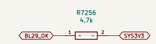


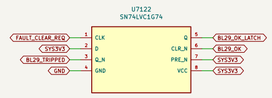














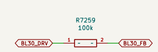


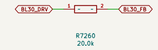


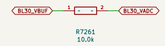
BL30

















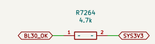


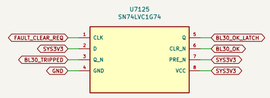
















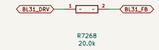


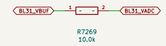
BL31













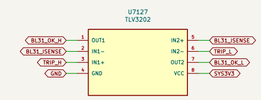






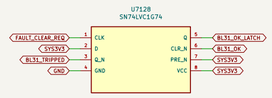


















MIRA

Sheet: /BL and NL analog channels/BL28 .. BL31 | drive, remote sense and signed current/
File: C_AFE_BL28_BL31.kicad_sch

Title: BL28 .. BL31 | drive, remote sense and signed current

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 83/73

NL0 .. NL3 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin. IADC measures current through the 5 ohm shunt.

U7129
OPA4192
NL0_DRV
NL0_FB
NL0_SET
P15
NL1_SET
NL1_FB
NL1_DRV
OUTA
INA−
INA+
VS+
INB+
INB−
OUTB
OUTC
INC−
INC+
VS−
IND+
IND−
OUTD
NL2_DRV
NL2_FB
NL2_SET
N15
NL3_SET
NL3_FB
NL3_DRV
C7365
100n
P15
GND
C7366
100n
N15
GND
U7130
OPA4192
NL0_VBUF
NL0_VBUF
NL0
P15
NL1
NL1_VBUF
NL1_VBUF
OUTA
INA−
INA+
VS+
INB+
INB−
OUTB
OUTC
INC−
INC+
VS−
IND+
IND−
OUTD
NL2_VBUF
NL2_VBUF
NL2
N15
NL3
NL3_VBUF
NL3_VBUF
C7367
100n
P15
GND
C7368
100n
N15
GND
U7131
ADG442
NL0_EN
NL0
NL0_S_MINUS
N15
GND
NL3_S_MINUS
NL3
NL3_EN
SEL1
D1
S1
VSS_NEG
GND
S4
D4
SEL4
SEL3
D3
S3
NC
VDD
S2
D2
SEL2
NL2_EN
NL2
NL2_S_MINUS
P15
NL1_S_MINUS
NL1
NL1_EN
C7369
100n
P15
GND
C7370
100n
N15
GND
U7132
TMUX6112
NL0_EN
NL0_FB_SW
NL0
N15
GND
NL3_FB_SW
NL3_EN
SEL1
D1
S1
VSS_NEG
GND
S4
D4
SEL4
SEL3
D3
S3
NC
VDD
S2
D2
SEL2
NL2_EN
NL2_FB_SW
NL2
P15
NL1
NL1_FB_SW
NL1_EN
C7371
100n
P15
GND
C7372
100n
N15
GND

NL0

R7273
5.00 0.1%
NL0_DRV
NL0_S_MINUS
R7274
1.00k
NL0_FB_SW
NL0_FB
R7275
100k
NL0_DRV
NL0_FB
R7276
20.0k
NL0_VBUF
NL0_VADC
R7277
10.0k
NL0_VADC
GND
R7278
47
NL0_ISENSE
NL0_IADC
R7279
47k
NL0_EN
GND
C7341
220p C0G
NL0_DRV
NL0_FB
C7342
220p C0G
NL0_VADC
GND
C7343
1n
NL0_IADC
GND
U7133
INA296A2
NL0_S_MINUS
GND
MID2V048
GND
IN−
GND
REF2
GND4
OUT
VS
REF1
IN+
NL0_ISENSE
SYS3V3
MID2V048
NL0_DRV
U7134
TLV3202
NL0_OK_H
NL0_ISENSE
TRIP_H
GND
OUT1
IN1−
IN1+
GND
IN2+
IN2−
OUT2
VCC
NL0_ISENSE
TRIP_L
NL0_OK_L
SYS3V3
D7033
BAT54A
NL0_OK_H
NL0_OK_L
K1
K2
A
NL0_OK
R7280
4.7k
NL0_OK
SYS3V3
U7135
SN74LVC1G74
FAULT_CLEAR_REQ
SYS3V3
NL0_TRIPPED
GND
CLK
D
Q_N
GND
Q
CLR_N
PRE_N
VCC
NL0_OK_LATCH
NL0_OK
SYS3V3
SYS3V3
C7344
100n
SYS3V3
GND
C7345
100n
SYS3V3
GND
C7346
100n
SYS3V3
GND

NL1

R7281
5.00 0.1%
NL1_DRV
NL1_S_MINUS
R7282
1.00k
NL1_FB_SW
NL1_FB
R7283
100k
NL1_DRV
NL1_FB
R7284
20.0k
NL1_VBUF
NL1_VADC
R7285
10.0k
NL1_VADC
GND
R7286
47
NL1_ISENSE
NL1_IADC
R7287
47k
NL1_EN
GND
C7347
220p C0G
NL1_DRV
NL1_FB
C7348
220p C0G
NL1_VADC
GND
C7349
1n
NL1_IADC
GND
U7136
INA296A2
NL1_S_MINUS
GND
MID2V048
GND
IN−
GND
REF2
GND4
OUT
VS
REF1
IN+
NL1_ISENSE
SYS3V3
MID2V048
NL1_DRV
U7137
TLV3202
NL1_OK_H
NL1_ISENSE
TRIP_H
GND
OUT1
IN1−
IN1+
GND
IN2+
IN2−
OUT2
VCC
NL1_ISENSE
TRIP_L
NL1_OK_L
SYS3V3
D7034
BAT54A
NL1_OK_H
NL1_OK_L
K1
K2
A
NL1_OK
R7288
4.7k
NL1_OK
SYS3V3
U7138
SN74LVC1G74
FAULT_CLEAR_REQ
SYS3V3
NL1_TRIPPED
GND
CLK
D
Q_N
GND
Q
CLR_N
PRE_N
VCC
NL1_OK_LATCH
NL1_OK
SYS3V3
SYS3V3
C7350
100n
SYS3V3
GND
C7351
100n
SYS3V3
GND
C7352
100n
SYS3V3
GND

NL2

R7289
5.00 0.1%
NL2_DRV
NL2_S_MINUS
R7290
1.00k
NL2_FB_SW
NL2_FB
R7291
100k
NL2_DRV
NL2_FB
R7292
20.0k
NL2_VBUF
NL2_VADC
R7293
10.0k
NL2_VADC
GND
R7294
47
NL2_ISENSE
NL2_IADC
R7295
47k
NL2_EN
GND
C7353
220p C0G
NL2_DRV
NL2_FB
C7354
220p C0G
NL2_VADC
GND
C7355
1n
NL2_IADC
GND
U7139
INA296A2
NL2_S_MINUS
GND
MID2V048
GND
IN−
GND
REF2
GND4
OUT
VS
REF1
IN+
NL2_ISENSE
SYS3V3
MID2V048
NL2_DRV
U7140
TLV3202
NL2_OK_H
NL2_ISENSE
TRIP_H
GND
OUT1
IN1−
IN1+
GND
IN2+
IN2−
OUT2
VCC
NL2_ISENSE
TRIP_L
NL2_OK_L
SYS3V3
D7035
BAT54A
NL2_OK_H
NL2_OK_L
K1
K2
A
NL2_OK
R7296
4.7k
NL2_OK
SYS3V3
U7141
SN74LVC1G74
FAULT_CLEAR_REQ
SYS3V3
NL2_TRIPPED
GND
CLK
D
Q_N
GND
Q
CLR_N
PRE_N
VCC
NL2_OK_LATCH
NL2_OK
SYS3V3
SYS3V3
C7356
100n
SYS3V3
GND
C7357
100n
SYS3V3
GND
C7358
100n
SYS3V3
GND

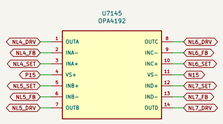
NL3

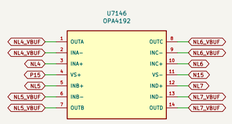
R7297
5.00 0.1%
NL3_DRV
NL3_S_MINUS
R7298
1.00k
NL3_FB_SW
NL3_FB
R7299
100k
NL3_DRV
NL3_FB
R7300
20.0k
NL3_VBUF
NL3_VADC
R7301
10.0k
NL3_VADC
GND
R7302
47
NL3_ISENSE
NL3_IADC
R7303
47k
NL3_EN
GND
C7359
220p C0G
NL3_DRV
NL3_FB
C7360
220p C0G
NL3_VADC
GND
C7361
1n
NL3_IADC
GND
U7142
INA296A2
NL3_S_MINUS
GND
MID2V048
GND
IN−
GND
REF2
GND4
OUT
VS
REF1
IN+
NL3_ISENSE
SYS3V3
MID2V048
NL3_DRV
U7143
TLV3202
NL3_OK_H
NL3_ISENSE
TRIP_H
GND
OUT1
IN1−
IN1+
GND
IN2+
IN2−
OUT2
VCC
NL3_ISENSE
TRIP_L
NL3_OK_L
SYS3V3
D7036
BAT54A
NL3_OK_H
NL3_OK_L
K1
K2
A
NL3_OK
R7304
4.7k
NL3_OK
SYS3V3
U7144
SN74LVC1G74
FAULT_CLEAR_REQ
SYS3V3
NL3_TRIPPED
GND
CLK
D
Q_N
GND
Q
CLR_N
PRE_N
VCC
NL3_OK_LATCH
NL3_OK
SYS3V3
SYS3V3
C7362
100n
SYS3V3
GND
C7363
100n
SYS3V3
GND
C7364
100n
SYS3V3
GND

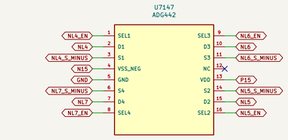
MIRA
Sheet: /BL and NL analog channels/NL0 .. NL3 | drive, remote sense and signed current/
File: C_AFE_NL0_NL3.kicad_sch
Title: NL0 .. NL3 | drive, remote sense and signed current
Size: A2
Date:
Rev: C-review
KiCad E.D.A. 10.0.8
Id: 84/73

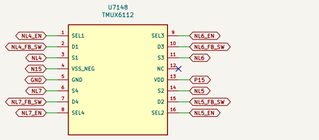
NL4 .. NL7 | drive, remote sense and signed current

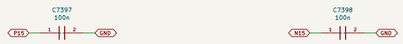
Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

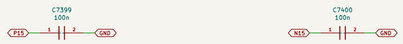


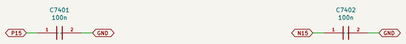


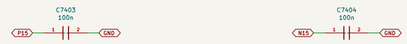




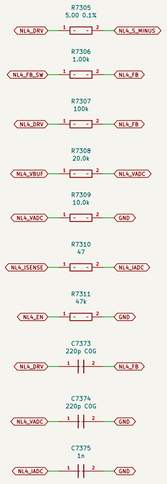


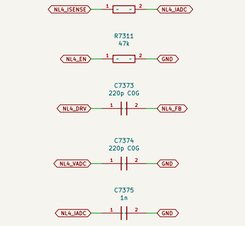


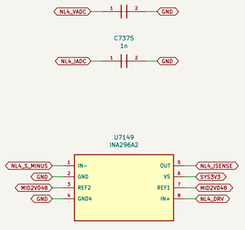




NL4

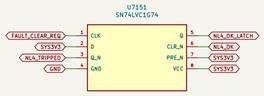






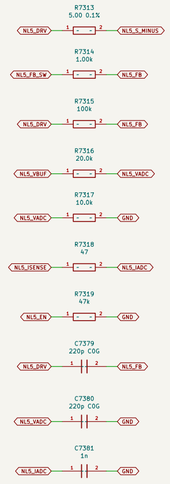


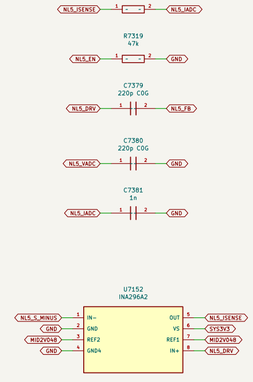






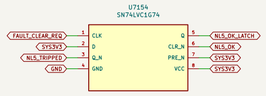
NL5







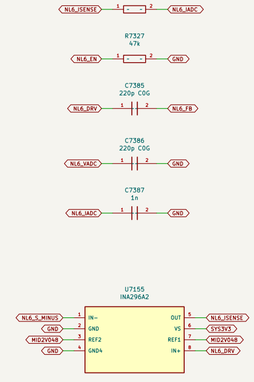






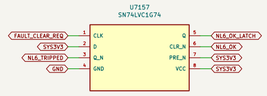
NL6





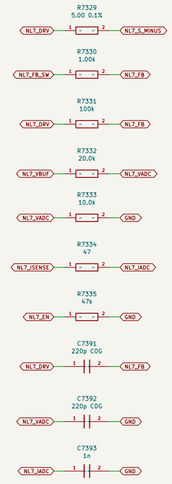


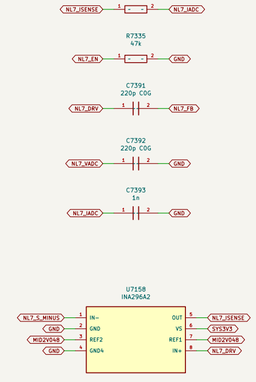




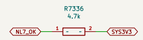


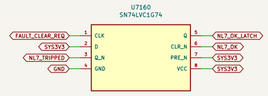
NL7













MIRA

Sheet: /BL and NL analog channels/NL4 .. NL7 | drive, remote sense and signed current/
File: C_AFE_NL4_NL7.kicad_sch

Title: NL4 .. NL7 | drive, remote sense and signed current

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 85/73

NL8 .. NL11 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

NL8

U7161
OPA4192
C7429
100n
C7430
100n
R7337
5.00 0.1%
R7338
1.00k
R7339
100k
R7340
20.0k
R7341
10.0k
R7342
47
R7343
47k
C7405
220p C0G
C7406
220p C0G
C7407
1n
U7165
INA296A2
U7166
TLV3202
D7041
BAT54A
R7344
4.7k
U7167
SN74LVC1G74
C7408
100n
C7409
100n
C7410
100n

NL9

U7162
OPA4192
C7431
100n
C7432
100n
R7345
5.00 0.1%
R7346
1.00k
R7347
100k
R7348
20.0k
R7349
10.0k
R7350
47
R7351
47k
C7411
220p C0G
C7412
220p C0G
C7413
1n
U7168
INA296A2
U7169
TLV3202
D7042
BAT54A
R7352
4.7k
U7170
SN74LVC1G74
C7414
100n
C7415
100n
C7416
100n

NL10

U7163
ADG442
C7433
100n
C7434
100n
R7353
5.00 0.1%
R7354
1.00k
R7355
100k
R7356
20.0k
R7357
10.0k
R7358
47
R7359
47k
C7417
220p C0G
C7418
220p C0G
C7419
1n
U7171
INA296A2
U7172
TLV3202
D7043
BAT54A
R7360
4.7k
U7173
SN74LVC1G74
C7420
100n
C7421
100n
C7422
100n

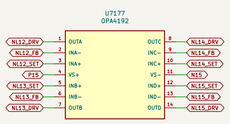
NL11

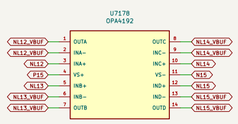
U7164
TMUX6112
C7435
100n
C7436
100n
R7361
5.00 0.1%
R7362
1.00k
R7363
100k
R7364
20.0k
R7365
10.0k
R7366
47
R7367
47k
C7423
220p C0G
C7424
220p C0G
C7425
1n
U7174
INA296A2
U7175
TLV3202
D7044
BAT54A
R7368
4.7k
U7176
SN74LVC1G74
C7426
100n
C7427
100n
C7428
100n

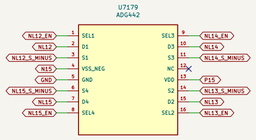
MIRA
Sheet: /BL and NL analog channels/NL8 .. NL11 | drive, remote sense and signed current/
File: C_AFE_NL8_NL11.kicad_sch
Title: NL8 .. NL11 | drive, remote sense and signed current
Size: A2
Date:
Rev: C-review
KiCad E.D.A. 10.0.8
Id: 86/73

NL12 .. NL15 | drive, remote sense and signed current

Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

















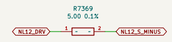








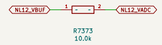
NL12











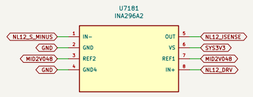






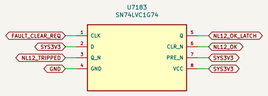


















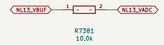
NL13











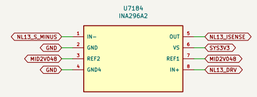


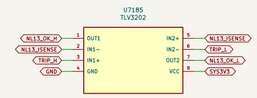




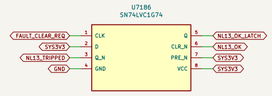
















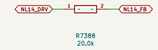


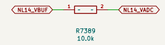
NL14





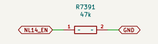






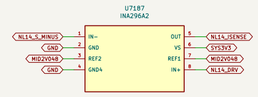


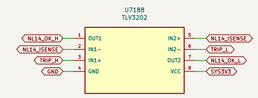




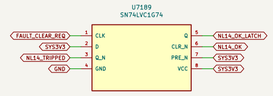


















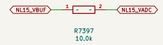
NL15





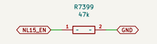






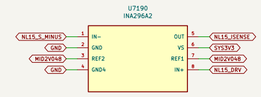


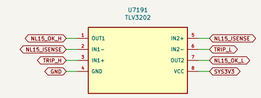




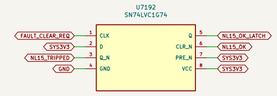














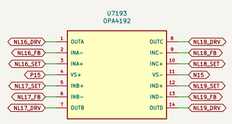




MIRA
Sheet: /BL and NL analog channels/NL12 .. NL15 | drive, remote sense and signed current/
File: C_AFE_NL12_NL15.kicad_sch
Title: NL12 .. NL15 | drive, remote sense and signed current
Size: A2 | Date: | **Rev: C-review**
KiCad E.D.A. 10.0.8 | Id: 87/73

NL16 .. NL19 | drive, remote sense and signed current

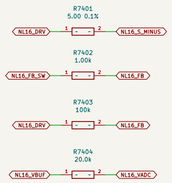
Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

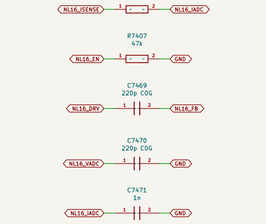






NL16



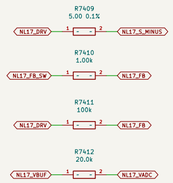


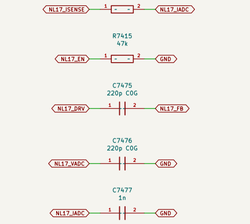


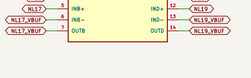




NL17



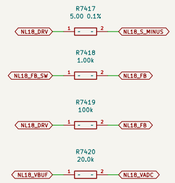


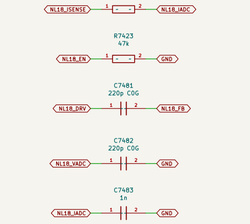


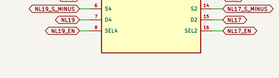




NL18



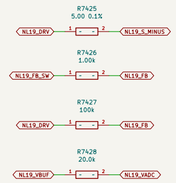


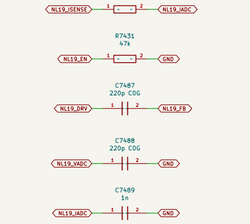






NL19











MIRA
Sheet: /BL and NL analog channels/NL16 .. NL19 | drive, remote sense and signed current/
File: C_AFE_NL16_NL19.kicad_sch

Title: NL16 .. NL19 | drive, remote sense and signed current

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 88/73

NL20 .. NL23 | drive, remote sense and signed current

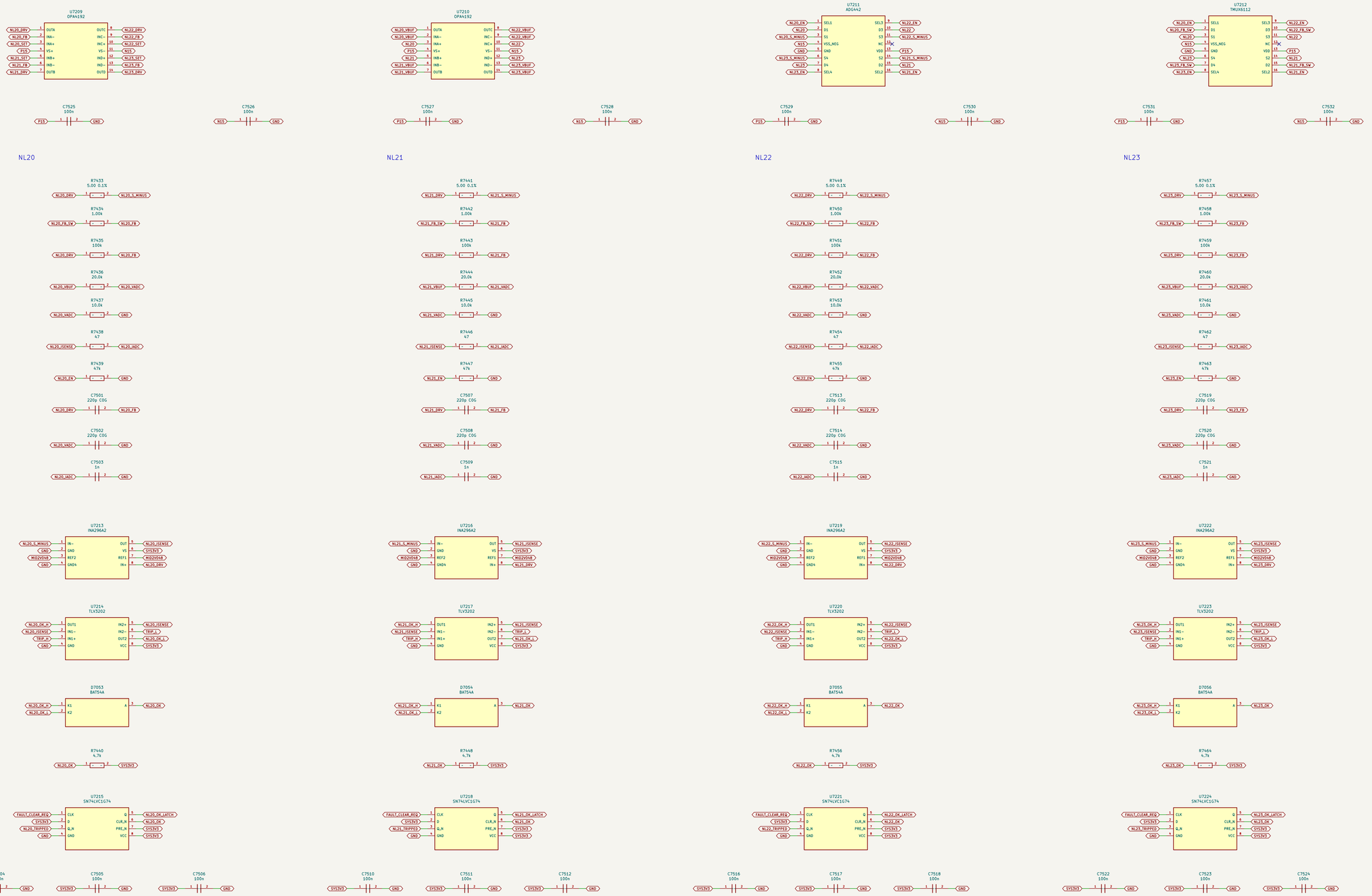
Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

NL20

NL21

NL22

NL23



MIRA

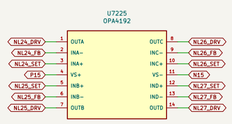
Sheet: /BL and NL analog channels/NL20 .. NL23 | drive, remote sense and signed current/
File: C_AFE_NL20_NL23.kicad_sch

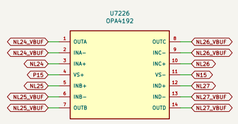
Title: NL20 .. NL23 | drive, remote sense and signed current

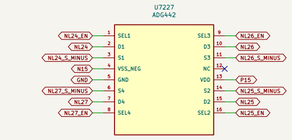
Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 89/73

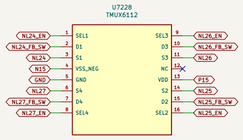
NL24 .. NL27 | drive, remote sense and signed current

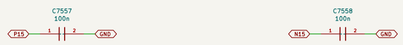
Each column below is one physical line. VADC is measured at the module pin, IADC measures current through the 5 ohm shunt.

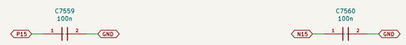


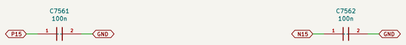


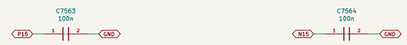




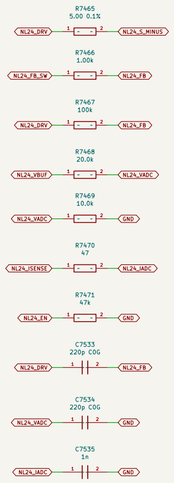


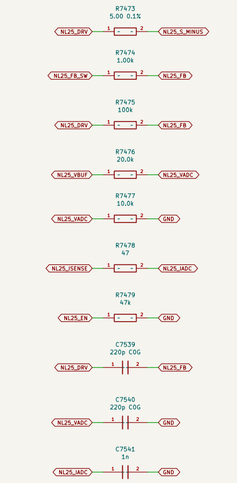


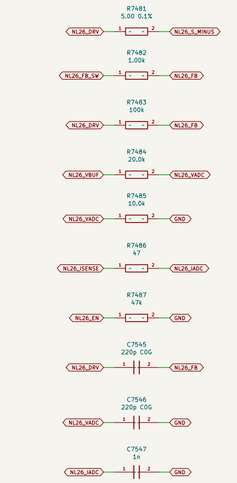


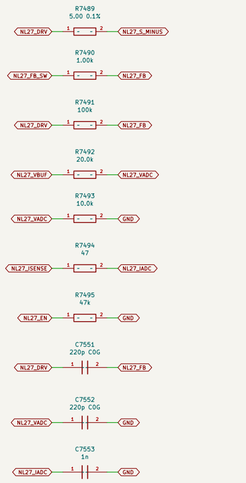


NL24



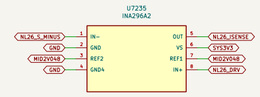


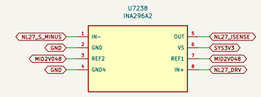


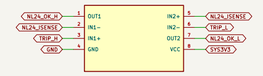


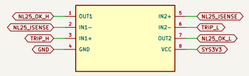


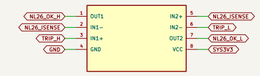


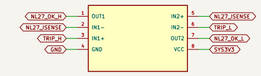






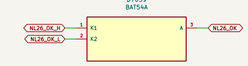








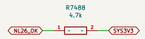


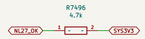


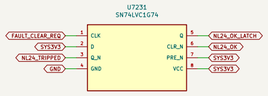


























MIRA

Sheet: /BL and NL analog channels/NL24 .. NL27 | drive, remote sense and signed current/
File: C_AFE_NL24_NL27.kicad_sch

Title: NL24 .. NL27 | drive, remote sense and signed current

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 90/73

NL28 .. NL31 | drive, remote sense and signed current

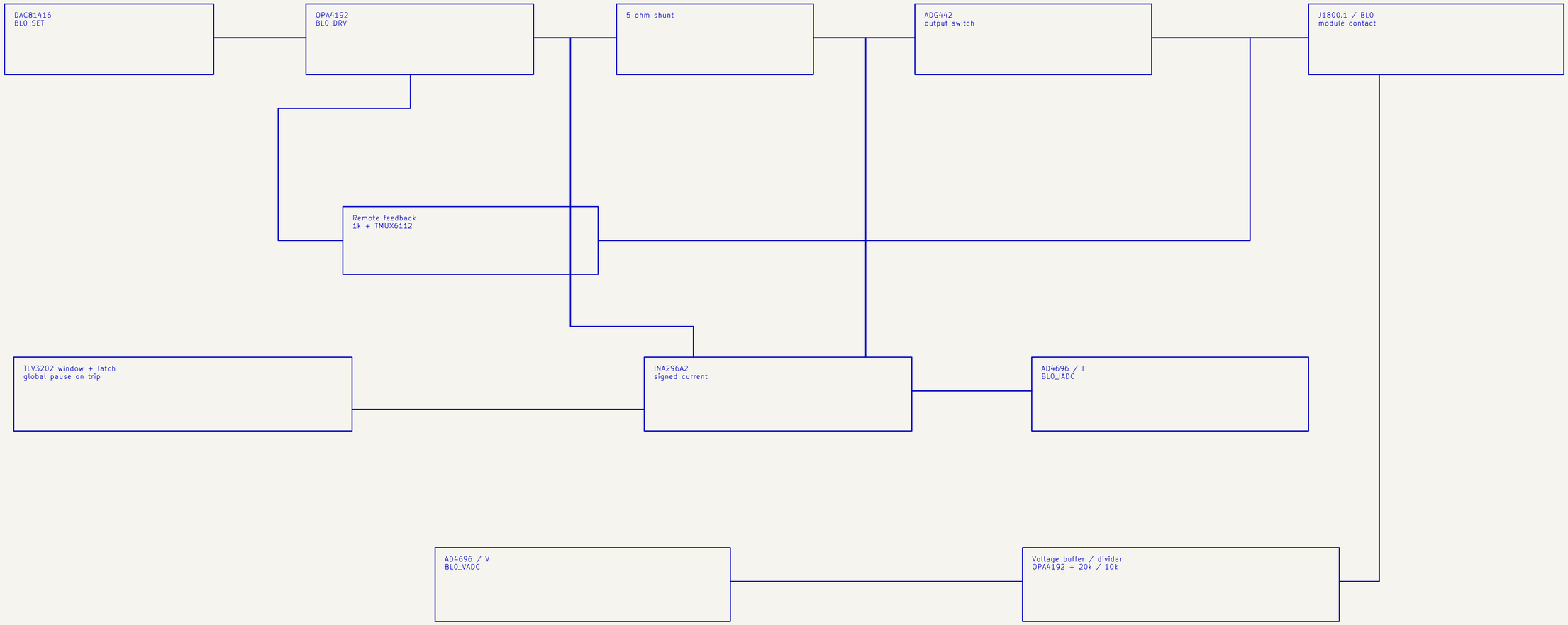
Each column below is one physical line. VADC is measured at the module pin. IADC measures current through the 5 ohm shunt.

U7241
OPA4192
C7589
100n
C7590
100n
U7242
OPA4192
C7591
100n
C7592
100n
U7243
ADG442
C7593
100n
C7594
100n
U7244
TMUX6112
C7595
100n
C7596
100n
NL28
R7497
5.00 0.1%
R7498
1.00k
R7499
100k
R7500
20.0k
R7501
10.0k
R7502
47
R7503
47k
C7565
220p C0G
C7566
220p C0G
C7567
1n
NL29
R7505
5.00 0.1%
R7506
1.00k
R7507
100k
R7508
20.0k
R7509
10.0k
R7510
47
R7511
47k
C7571
220p C0G
C7572
220p C0G
C7573
1n
NL30
R7513
5.00 0.1%
R7514
1.00k
R7515
100k
R7516
20.0k
R7517
10.0k
R7518
47
R7519
47k
C7577
220p C0G
C7578
220p C0G
C7579
1n
NL31
R7521
5.00 0.1%
R7522
1.00k
R7523
100k
R7524
20.0k
R7525
10.0k
R7526
47
R7527
47k
C7583
220p C0G
C7584
220p C0G
C7585
1n
U7245
INA296A2
U7246
TLV3202
D7061
BAT54A
R7504
4.7k
U7247
SN74LVC1G74
U7248
INA296A2
U7249
TLV3202
D7062
BAT54A
R7512
4.7k
U7250
SN74LVC1G74
U7251
INA296A2
U7252
TLV3202
D7063
BAT54A
R7520
4.7k
U7253
SN74LVC1G74
U7254
INA296A2
U7255
TLV3202
D7064
BAT54A
R7528
4.7k
U7256
SN74LVC1G74
C7568
100n
C7569
100n
C7570
100n
C7574
100n
C7575
100n
C7576
100n
C7580
100n
C7581
100n
C7582
100n
C7586
100n
C7587
100n
C7588
100n

MIRA
Sheet: /BL and NL analog channels/NL28 .. NL31 | drive, remote sense and signed current/
File: C_AFE_NL28_NL31.kicad_sch
Title: NL28 .. NL31 | drive, remote sense and signed current
Size: A2
Date:
Rev: C-review
KiCad E.D.A. 10.0.8
Id: 91/73

How to read a BL or NL channel

Example BL0. Identical topology on all 64 lines. The nets shown here are implemented on the AFE sheets.



1 The local 100k / 220p feedback remains connected when the remote loop opens.

2 Nominal current transfer: 2.048 V + 100 V/A x I. Positive current is delivered into the matrix.

3 VADC = V(BL0)/3. Differential cell voltage uses the measured BL and NL values.

4 The forming return uses its own shunt. During a fault, conversions can continue with outputs disabled.

MIRA
Sheet: /BL and NL analog channels/How to read a BL or NL channel/
File: C_ChannelGuide.kicad_sch
Title: How to read a BL or NL channel
Size: A2 | Date: | Rev: C-review
KiCad E.D.A. 10.0.8 | Id: 92/73

Word line drivers

WL0 .. WL3 | switched VWL and active discharge

WL4 .. WL7 | switched VWL and active discharge

WL8 .. WL11 | switched VWL and active discharge

WL12 .. WL15 | switched VWL and active discharge

WL16 .. WL19 | switched VWL and active discharge

WL20 .. WL23 | switched VWL and active discharge

WL24 .. WL27 | switched VWL and active discharge

WL28 .. WL31 | switched VWL and active discharge

WL pre-regulator and enable

Sheet: /Word line drivers/
File: C_GROUP_WORDLINES.kicad_sch

Title: Word line drivers

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 93/73

WL0 .. WL3 | switched VWL and active discharge

Conditional turn-on in each direction. RC plus Schmitt stage sets dead time. Fast deassertion of each gate is independent of the RC.

U7273
TMUX6212
WL0_UP
WL0_UP_RAW
VWL_ON
GND
GND
VWL_ON
WL3_UP_RAW
WL3_UP
SEL1
D1
S1
VSS_NEG
GND
S4
D4
SEL4
SEL3
D3
S3
NC
VDD
S2
D2
SEL2
WL2_UP
WL2_UP_RAW
VWL_ON
VWL_PRE12
VWL_ON
WL1_UP_RAW
WL1_UP

U7274
SN74LVC125A
GND
WL0_DN
WL0_OUT
GND
WL1_DN
WL1_OUT
GND
OE1_N
A1
Y1
OE2_N
A2
Y2
GND
Y3
A3
OE3_N
Y4
A4
OE4_N
VCC
WL2_OUT
WL2_DN
GND
WL3_OUT
WL3_DN
GND
SYS3V3

C7614
2.2u effective
VWL_ON
GND

C7615
100n
VWL_PRE12
GND

C7616
100n
SYS3V3
GND

U7331A
SN74LVC08A
WL_REQ0
RUN_NORMAL
A
B
Y
WL0_NORMAL

U7331C
SN74LVC08A
WL_REQ1
RUN_NORMAL
WL1_NORMAL

U7332A
SN74LVC08A
WL_REQ2
RUN_NORMAL
WL2_NORMAL

U7332C
SN74LVC08A
WL_REQ3
RUN_NORMAL
WL3_NORMAL

U7333A
SN74LVC32A
WL0_NORMAL
WL0_FORM
WL0_WANT

U7333B
SN74LVC32A
WL1_NORMAL
WL1_FORM
WL1_WANT

U7333C
SN74LVC32A
WL2_NORMAL
WL2_FORM
WL2_WANT

U7333D
SN74LVC32A
WL3_NORMAL
WL3_FORM
WL3_WANT

R7537
4.7k
WL0_WANT
WL0_RC

R7542
4.7k
WL1_WANT
WL1_RC

R7547
4.7k
WL2_WANT
WL2_RC

R7552
4.7k
WL3_WANT
WL3_RC

C7617
100p C0G
WL0_RC
GND

C7618
100p C0G
WL1_RC
GND

C7619
100p C0G
WL2_RC
GND

C7620
100p C0G
WL3_RC
GND

U7334A
SN74LVC14A
WL0_RC
WL0_LATE_N

U7334C
SN74LVC14A
WL1_RC
WL1_LATE_N

U7334E
SN74LVC14A
WL2_RC
WL2_LATE_N

U7335A
SN74LVC14A
WL3_RC
WL3_LATE_N

U7334B
SN74LVC14A
WL0_LATE_N
WL0_LATE

U7334D
SN74LVC14A
WL1_LATE_N
WL1_LATE

U7334F
SN74LVC14A
WL2_LATE_N
WL2_LATE

U7335B
SN74LVC14A
WL3_LATE_N
WL3_LATE

U7331B
SN74LVC08A
WL0_WANT
WL0_LATE
WL0_UP

U7331D
SN74LVC08A
WL1_WANT
WL1_LATE
WL1_UP

U7332B
SN74LVC08A
WL2_WANT
WL2_LATE
WL2_UP

U7332D
SN74LVC08A
WL3_WANT
WL3_LATE
WL3_UP

U7336A
SN74LVC02A
WL0_WANT
WL0_LATE
WL0_DN

U7336B
SN74LVC02A
WL1_WANT
WL1_LATE
WL1_DN

U7336C
SN74LVC02A
WL2_WANT
WL2_LATE
WL2_DN

U7336D
SN74LVC02A
WL3_WANT
WL3_LATE
WL3_DN

R7538
120
WL0_UP_RAW
WL0

R7543
120
WL1_UP_RAW
WL1

R7548
120
WL2_UP_RAW
WL2

R7553
120
WL3_UP_RAW
WL3

R7539
27
WL0
WL0_DRAIN

R7544
27
WL1
WL1_DRAIN

R7549
27
WL2
WL2_DRAIN

R7554
27
WL3
WL3_DRAIN

R7540
180
WL0_OUT
WL0_GATE

R7545
180
WL1_OUT
WL1_GATE

R7550
180
WL2_OUT
WL2_GATE

R7555
180
WL3_OUT
WL3_GATE

R7541
47k
WL0_GATE
GND

R7546
47k
WL1_GATE
GND

R7551
47k
WL2_GATE
GND

R7556
47k
WL3_GATE
GND

Q7001
PMV20XNEA
WL0_GATE
GND
G
S
D
WL0_DRAIN

Q7002
PMV20XNEA
WL1_GATE
GND
WL1_DRAIN

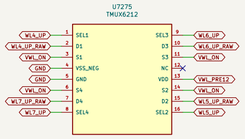
Q7003
PMV20XNEA
WL2_GATE
GND
WL2_DRAIN

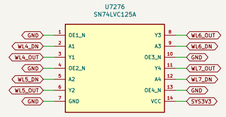
Q7004
PMV20XNEA
WL3_GATE
GND
WL3_DRAIN

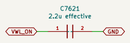
MIRA
Sheet: /Word line drivers/WL0 .. WL3 | switched VWL and active discharge/
File: C_Rows0.kicad_sch
Title: WL0 .. WL3 | switched VWL and active discharge
Size: A2
Date:
Rev: C-review
KiCad E.D.A. 10.0.8
Id: 94/73

WL4 .. WL7 | switched VWL and active discharge

Conditional turn-on in each direction. RC plus Schmitt stage sets dead time. Fast deassertion of each gate is independent of the RC.











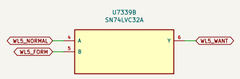






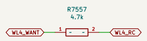


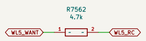


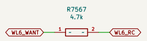


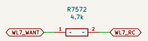














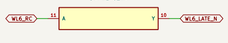


































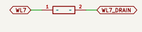


























MIRA

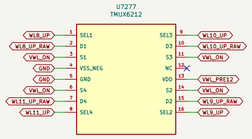
Sheet: /Word line drivers/WL4 .. WL7 | switched VWL and active discharge/
File: C_Rows1.kicad_sch

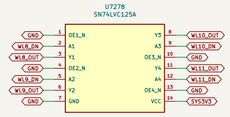
Title: WL4 .. WL7 | switched VWL and active discharge

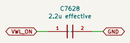
Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 95/73

WL8 .. WL11 | switched VWL and active discharge

Conditional turn-on in each direction. RC plus Schmitt stage sets dead time. Fast deassertion of each gate is independent of the RC.





















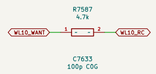




































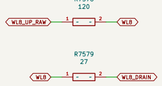


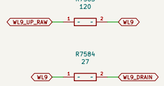


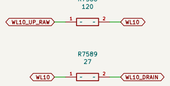


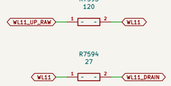


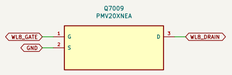


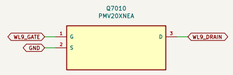


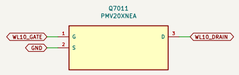


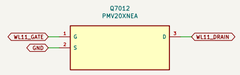












MIRA

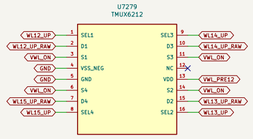
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File: C_Rows2.kicad_sch

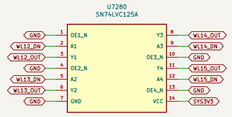
Title: WL8 .. WL11 | switched VWL and active discharge

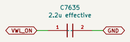
Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 96/73

WL12 .. WL15 | switched VWL and active discharge

Conditional turn-on in each direction. RC plus Schmitt stage sets dead time. Fast deassertion of each gate is independent of the RC.



















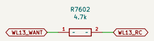




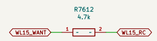
































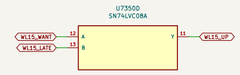


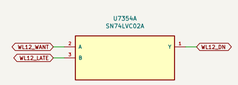








































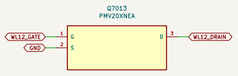


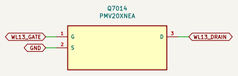


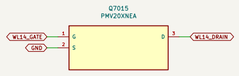


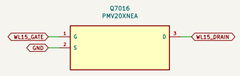












MIRA

Sheet: /Word line drivers/WL12 .. WL15 | switched VWL and active discharge/
File: C_Rows3.kicad_sch

Title: WL12 .. WL15 | switched VWL and active discharge

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 97/73

WL16 .. WL19 | switched VWL and active discharge

Conditional turn-on in each direction. RC plus Schmitt stage sets dead time. Fast deassertion of each gate is independent of the RC.

U7281
TMUX6212
WL16_UP
WL16_UP_RAW
VWL_ON
GND
GND
VWL_ON
WL19_UP_RAW
WL19_UP
SEL1
D1
S1
VSS_NEG
GND
S4
D4
SEL4
SEL3
D3
S3
NC
VDD
S2
D2
SEL2
WL18_UP
WL18_UP_RAW
VWL_ON
VWL_PRE12
VWL_ON
WL17_UP_RAW
WL17_UP

U7282
SN74LVC125A
GND
WL16_DN
WL16_OUT
GND
WL17_DN
WL17_OUT
GND
OE1_N
A1
Y1
OE2_N
A2
Y2
GND
Y3
A3
OE3_N
Y4
A4
OE4_N
VCC
WL18_OUT
WL18_DN
GND
WL19_OUT
WL19_DN
GND
SYS3V3

C7642
2.2u effective
VWL_ON
GND

C7643
100n
VWL_PRE12
GND

C7644
100n
SYS3V3
GND

U7355A
SN74LVC08A
WL_REQ16
RUN_NORMAL
A
B
Y
WL16_NORMAL

U7355C
SN74LVC08A
WL_REQ17
RUN_NORMAL
A
B
Y
WL17_NORMAL

U7356A
SN74LVC08A
WL_REQ18
RUN_NORMAL
A
B
Y
WL18_NORMAL

U7356C
SN74LVC08A
WL_REQ19
RUN_NORMAL
A
B
Y
WL19_NORMAL

U7357A
SN74LVC32A
WL16_NORMAL
WL16_FORM
A
B
Y
WL16_WANT

U7357B
SN74LVC32A
WL17_NORMAL
WL17_FORM
A
B
Y
WL17_WANT

U7357C
SN74LVC32A
WL18_NORMAL
WL18_FORM
A
B
Y
WL18_WANT

U7357D
SN74LVC32A
WL19_NORMAL
WL19_FORM
A
B
Y
WL19_WANT

R7617
4.7k
WL16_WANT
WL16_RC

R7622
4.7k
WL17_WANT
WL17_RC

R7627
4.7k
WL18_WANT
WL18_RC

R7632
4.7k
WL19_WANT
WL19_RC

C7645
100p C0G
WL16_RC
GND

C7646
100p C0G
WL17_RC
GND

C7647
100p C0G
WL18_RC
GND

C7648
100p C0G
WL19_RC
GND

U7358A
SN74LVC14A
WL16_RC
A
Y
WL16_LATE_N

U7358C
SN74LVC14A
WL17_RC
A
Y
WL17_LATE_N

U7358E
SN74LVC14A
WL18_RC
A
Y
WL18_LATE_N

U7359A
SN74LVC14A
WL19_RC
A
Y
WL19_LATE_N

U7358B
SN74LVC14A
WL16_LATE_N
A
Y
WL16_LATE

U7358D
SN74LVC14A
WL17_LATE_N
A
Y
WL17_LATE

U7358F
SN74LVC14A
WL18_LATE_N
A
Y
WL18_LATE

U7359B
SN74LVC14A
WL19_LATE_N
A
Y
WL19_LATE

U7355B
SN74LVC08A
WL16_WANT
WL16_LATE
A
B
Y
WL16_UP

U7355D
SN74LVC08A
WL17_WANT
WL17_LATE
A
B
Y
WL17_UP

U7356B
SN74LVC08A
WL18_WANT
WL18_LATE
A
B
Y
WL18_UP

U7356D
SN74LVC08A
WL19_WANT
WL19_LATE
A
B
Y
WL19_UP

U7360A
SN74LVC02A
WL16_WANT
WL16_LATE
A
B
Y
WL16_DN

U7360B
SN74LVC02A
WL17_WANT
WL17_LATE
A
B
Y
WL17_DN

U7360C
SN74LVC02A
WL18_WANT
WL18_LATE
A
B
Y
WL18_DN

U7360D
SN74LVC02A
WL19_WANT
WL19_LATE
A
B
Y
WL19_DN

R7618
120
WL16_UP_RAW
WL16

R7623
120
WL17_UP_RAW
WL17

R7628
120
WL18_UP_RAW
WL18

R7633
120
WL19_UP_RAW
WL19

R7619
27
WL16
WL16_DRAIN

R7624
27
WL17
WL17_DRAIN

R7629
27
WL18
WL18_DRAIN

R7634
27
WL19
WL19_DRAIN

R7620
180
WL16_OUT
WL16_GATE

R7625
180
WL17_OUT
WL17_GATE

R7630
180
WL18_OUT
WL18_GATE

R7635
180
WL19_OUT
WL19_GATE

R7621
47k
WL16_GATE
GND

R7626
47k
WL17_GATE
GND

R7631
47k
WL18_GATE
GND

R7636
47k
WL19_GATE
GND

Q7017
PMV20XNEA
WL16_GATE
GND
G
S
D
WL16_DRAIN

Q7018
PMV20XNEA
WL17_GATE
GND
G
S
D
WL17_DRAIN

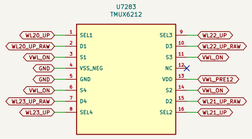
Q7019
PMV20XNEA
WL18_GATE
GND
G
S
D
WL18_DRAIN

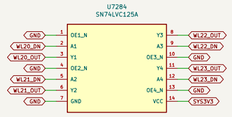
Q7020
PMV20XNEA
WL19_GATE
GND
G
S
D
WL19_DRAIN

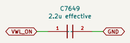
MIRA
Sheet: /Word line drivers/WL16 .. WL19 | switched VWL and active discharge/
File: C_Rows4.kicad_sch
Title: WL16 .. WL19 | switched VWL and active discharge
Size: A2
Date:
Rev: C-review
KiCad E.D.A. 10.0.8
Id: 98/73

WL20 .. WL23 | switched VWL and active discharge

Conditional turn-on in each direction. RC plus Schmitt stage sets dead time. Fast deassertion of each gate is independent of the RC.























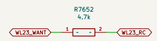










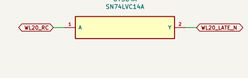


















































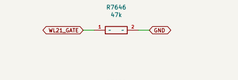


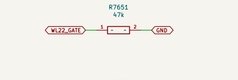


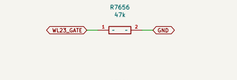












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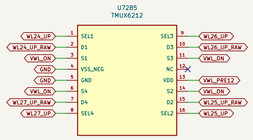
Sheet: /Word line drivers/WL20 .. WL23 | switched VWL and active discharge/
File: C_Rows5.kicad_sch

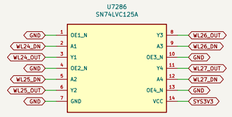
Title: WL20 .. WL23 | switched VWL and active discharge

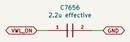
Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 99/73

WL24 .. WL27 | switched VWL and active discharge

Conditional turn-on in each direction. RC plus Schmitt stage sets dead time. Fast deassertion of each gate is independent of the RC.

















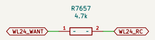


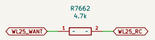






































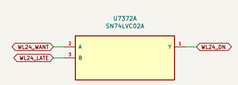






















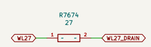


















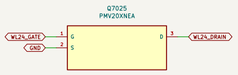


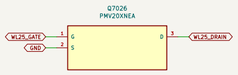


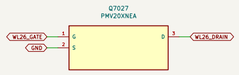


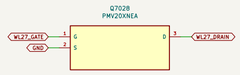












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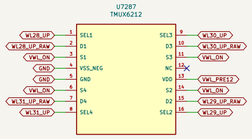
Sheet: /Word line drivers/WL24 .. WL27 | switched VWL and active discharge/
File: C_Rows6.kicad_sch

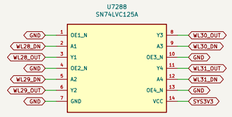
Title: WL24 .. WL27 | switched VWL and active discharge

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 100/73

WL28 .. WL31 | switched VWL and active discharge

Conditional turn-on in each direction. RC plus Schmitt stage sets dead time. Fast deassertion of each gate is independent of the RC.



















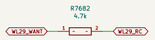






























































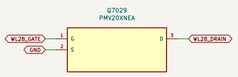




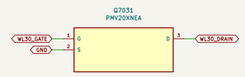














MIRA

Sheet: /Word line drivers/WL28 .. WL31 | switched VWL and active discharge/
File: C_Rows7.kicad_sch

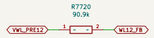
Title: WL28 .. WL31 | switched VWL and active discharge

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 101/73

WL pre-regulator and enable

15 V analog rail to nominal 12.0 V. Verify total 32-row capacitive load, input sag, startup and thermal headroom.





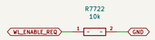












MIRA
Sheet: /Word line drivers/WL pre-regulator and enable/
File: C_WL12.kicad_sch
Title: WL pre-regulator and enable
Size: A2 | Date: | **Rev: C-review**
KiCad E.D.A. 10.0.8 | Id: 102/73

Interlocks and forming

Drive masks and fault memory | BL0 .. BL7

Drive masks and fault memory | BL8 .. BL15

Drive masks and fault memory | BL16 .. BL23

Drive masks and fault memory | BL24 .. BL31

Drive masks and fault memory | NL0 .. NL7

Drive masks and fault memory | NL8 .. NL15

Drive masks and fault memory | NL16 .. NL23

Drive masks and fault memory | NL24 .. NL31

Global fault latch and mode interlocks

Forming | one BL, one NL, one physical WL

Service DAC, reference, WL supply and forming servo

Mode-change blanking and reset interfaces

Logic supply and spare gates | 1

Spare gates | 1

Logic supply and spare gates | 2

Logic supply and spare gates | 3

Spare gates | 2

Forming address and polarity lock

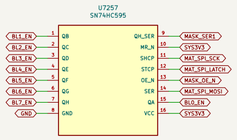
Sheet: /Interlocks and forming/
File: C_GROUP_SAFETY.kicad_sch

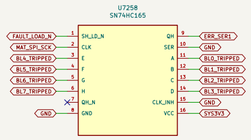
Title: Interlocks and forming

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 103/73

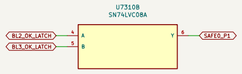
Drive masks and fault memory | BL0 .. BL7

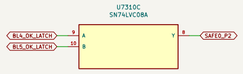
The storage register enables individual lines. Latched per-line faults feed a hardware AND tree and can be read serially.







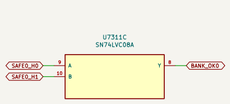
















MIRA

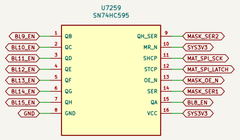
Sheet: /Interlocks and forming/Drive masks and fault memory | BL0 .. BL7/
File: C_Mask0.kicad_sch

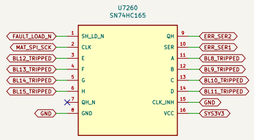
Title: Drive masks and fault memory | BL0 .. BL7

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 104/73

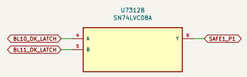
Drive masks and fault memory | BL8 .. BL15

The storage register enables individual lines. Latched per-line faults feed a hardware AND tree and can be read serially.























MIRA

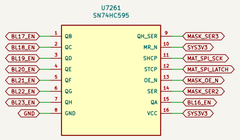
Sheet: /Interlocks and forming/Drive masks and fault memory | BL8 .. BL15/
File: C_Mask1.kicad_sch

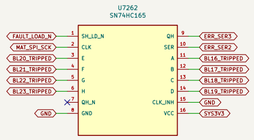
Title: Drive masks and fault memory | BL8 .. BL15

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 105/73

Drive masks and fault memory | BL16 .. BL23

The storage register enables individual lines. Latched per-line faults feed a hardware AND tree and can be read serially.





















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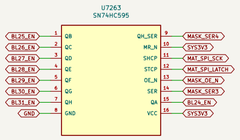
Sheet: /Interlocks and forming/Drive masks and fault memory | BL16 .. BL23/
File: C_Mask2.kicad_sch

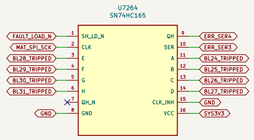
Title: Drive masks and fault memory | BL16 .. BL23

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 106/73

Drive masks and fault memory | BL24 .. BL31

The storage register enables individual lines. Latched per-line faults feed a hardware AND tree and can be read serially.

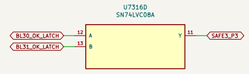






















MIRA

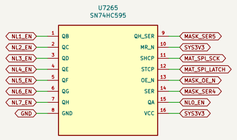
Sheet: /Interlocks and forming/Drive masks and fault memory | BL24 .. BL31/
File: C_Mask3.kicad_sch

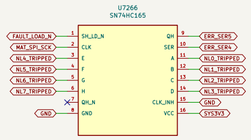
Title: Drive masks and fault memory | BL24 .. BL31

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 107/73

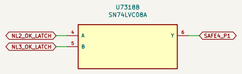
Drive masks and fault memory | NL0 .. NL7

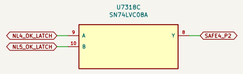
The storage register enables individual lines. Latched per-line faults feed a hardware AND tree and can be read serially.







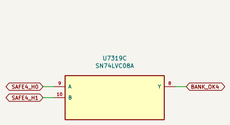
















MIRA

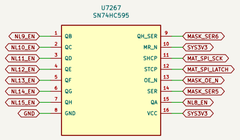
Sheet: /Interlocks and forming/Drive masks and fault memory | NL0 .. NL7/
File: C_Mask4.kicad_sch

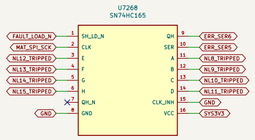
Title: Drive masks and fault memory | NL0 .. NL7

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 108/73

Drive masks and fault memory | NL8 .. NL15

The storage register enables individual lines. Latched per-line faults feed a hardware AND tree and can be read serially.

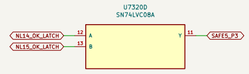






















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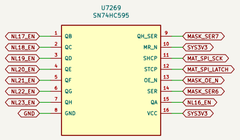
Sheet: /Interlocks and forming/Drive masks and fault memory | NL8 .. NL15/
File: C_Mask5.kicad_sch

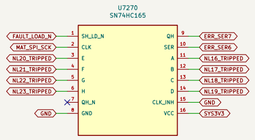
Title: Drive masks and fault memory | NL8 .. NL15

Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 109/73

Drive masks and fault memory | NL16 .. NL23

The storage register enables individual lines. Latched per-line faults feed a hardware AND tree and can be read serially.

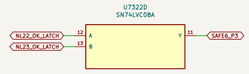






















MIRA

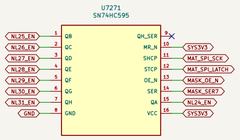
Sheet: /Interlocks and forming/Drive masks and fault memory | NL16 .. NL23/
File: C_Mask6.kicad_sch

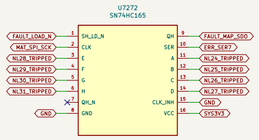
Title: Drive masks and fault memory | NL16 .. NL23

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 110/73

Drive masks and fault memory | NL24 .. NL31

The storage register enables individual lines. Latched per-line faults feed a hardware AND tree and can be read serially.























MIRA

Sheet: /Interlocks and forming/Drive masks and fault memory | NL24 .. NL31/
File: C_Mask7.kicad_sch

Title: Drive masks and fault memory | NL24 .. NL31

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 111/73

Global fault latch and mode interlocks

Power permit, software run request and hardware protection are in series. ADC sampling stays active after a trip.











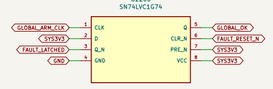














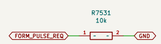
























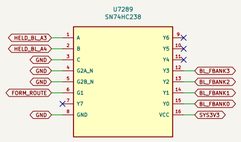


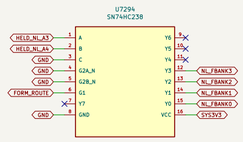
Recovery: keep RUN_REQ low, clear local memories, confirm status, then pulse GLOBAL_ARM_CLK. Open inhibit, lost permit or overcurrent forces a fresh arm.

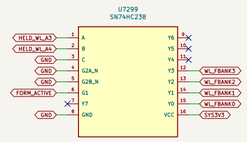
MIRA	
Sheet: /Interlocks and forming/Global fault latch and mode interlocks/	
File: C_Global.kicad_sch	
Title: Global fault latch and mode interlocks	
Size: A2	Date:
KiCad E.D.A. 10.0.8	**Rev: C-review**
	Id: 112/73

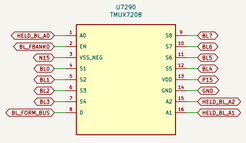
Forming | one BL, one NL, one physical WL

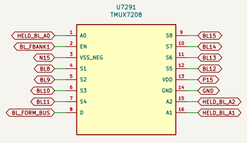
Each 32:1 line selector uses four TSSOP 8:1 analog switches. Address is held stable while disconnected. WL follows a separately programmed five-bit index.

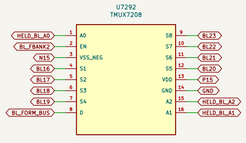


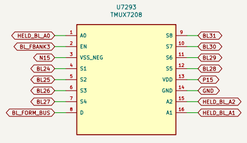


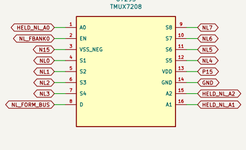


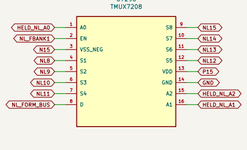


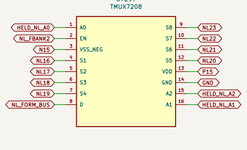


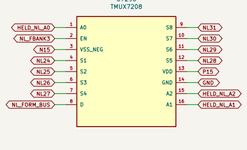




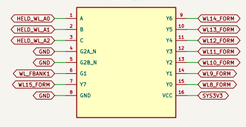




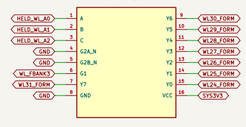


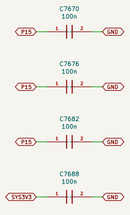


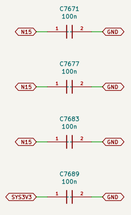


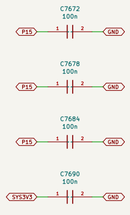


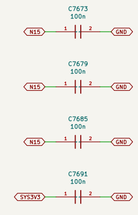


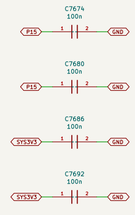


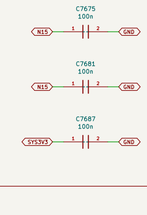












MIRA

Sheet: /Interlocks and forming/Forming | one BL, one NL, one physical WL/
File: C_FormSelect.kicad_sch

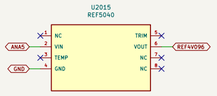
Title: Forming | one BL, one NL, one physical WL

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 113/73

Service DAC, reference, WL supply and forming servo

Program thresholds with RUN_REQ low. 100 ohm return shunt with limited reference sets current compliance in either routed polarity.





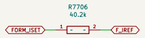














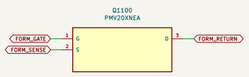


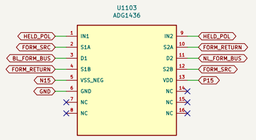






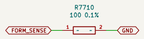
















Hardware trip -11.35 mA. Actual cell voltage is reduced by series switches and the return shunt. Measure differential BL−NL during forming.













































MIRA

Sheet: /Interlocks and forming/Service DAC, reference, WL supply and forming servo/

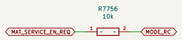
File: C_Service.kicad_sch

Title: Service DAC, reference, WL supply and forming servo

Size: A2	Date:	**Rev: C−review**
KiCad E.D.A. 10.0.8		Id: 114/73

Mode-change blanking and reset interfaces

Inhibit driver families while mode is changing. The flip-flops still require separate clear and arm pulses.















MIRA

Sheet: /Interlocks and forming/Mode-change blanking and reset interfaces/
File: C_Mode.kicad_sch

Title: Mode-change blanking and reset interfaces

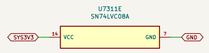
Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 115/73

Logic supply and spare gates | 1

One 100 nF decoupler per package. Grounds on unused inputs are intentional.



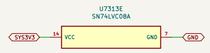




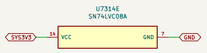




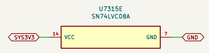










































































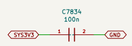












MIRA

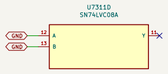
Sheet: /Interlocks and forming/Logic supply and spare gates | 1/
File: C_LogicPower0.kicad_sch

Title: Logic supply and spare gates | 1

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 116/73

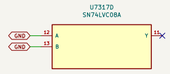
Spare gates | 1

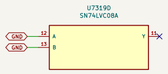
Unused CMOS inputs have defined levels.



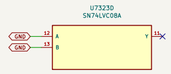


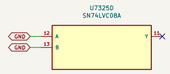


















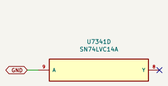


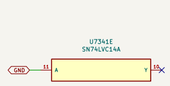


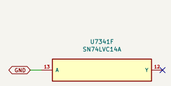


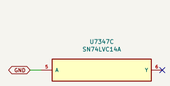




































MIRA

Sheet: /Interlocks and forming/Spare gates | 1/
File: C_SpareLogic0.kicad_sch

Title: Spare gates | 1

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 117/73

Logic supply and spare gates | 2

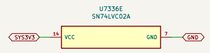
One 100 nF decoupler per package. Grounds on unused inputs are intentional.







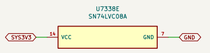




































































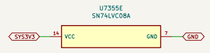






















MIRA

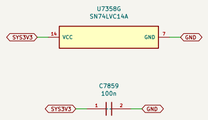
Sheet: /Interlocks and forming/Logic supply and spare gates | 2/
File: C_LogicPower1.kicad_sch

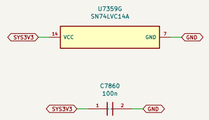
Title: Logic supply and spare gates | 2

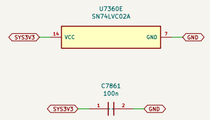
Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 118/73

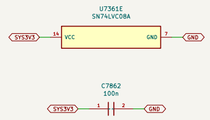
Logic supply and spare gates | 3

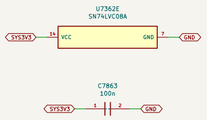
One 100 nF decoupler per package. Grounds on unused inputs are intentional.

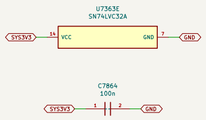


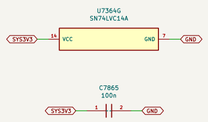


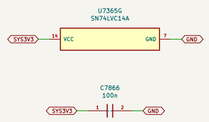


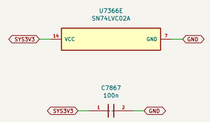


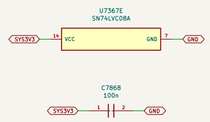


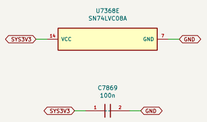


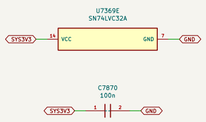


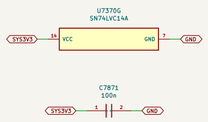


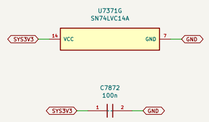


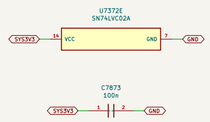


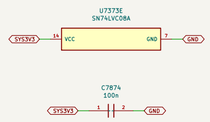


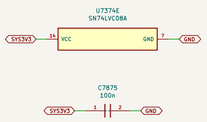


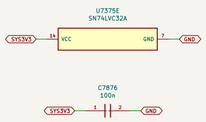


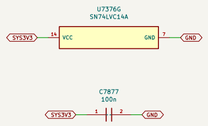


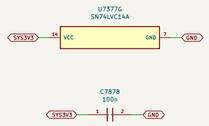


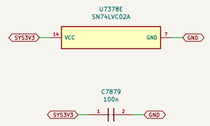


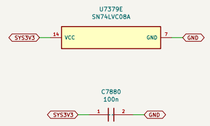


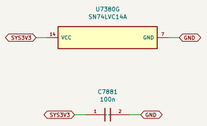


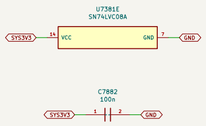












MIRA

Sheet: /Interlocks and forming/Logic supply and spare gates | 3/
File: C_LogicPower2.kicad_sch

Title: Logic supply and spare gates | 3

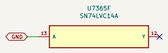
Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 119/73

Spare gates | 2

Unused CMOS inputs have defined levels.



















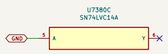














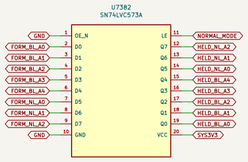
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File: C_SpareLogic1.kicad_sch

Title: Spare gates | 2

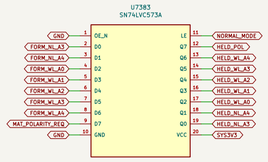
Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 120/73

Forming address and polarity lock

A transparent latch follows each setting with service mode off. The latches close when service is requested, ahead of the delayed switch enable.









MIRA
Sheet: /Interlocks and forming/Forming address and polarity lock/
File: C_FormAddress.kicad_sch
Title: Forming address and polarity lock
Size: A2 | Date: | **Rev: C-review**
KiCad E.D.A. 10.0.8 | Id: 121/73

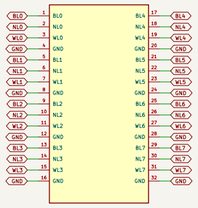
Passive module interface

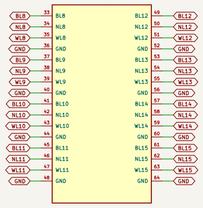
Passive module | 140 contacts

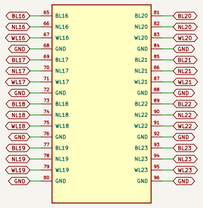
Sheet: /Passive module interface/		
File: C_GROUP_MODULE.kicad_sch		
Title: Passive module interface		
Size: A2	Date:	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 122/73

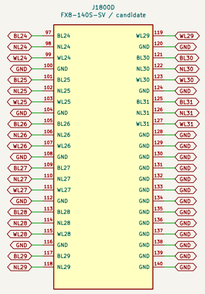
Passive module | 140 contacts

Fixed logical pinout. With a ground return for each three signals. Hirose FX8 mechanical candidate. Last 12 contacts are additional grounds.









Mating module contains only routing, grounds and the matrix. Same contacts for forming. 1R modules leave WL open. Pin numbering is viewed from the Control mating face.

MIRA

Sheet: /Passive module interface/Passive module | 140 contacts/
File: C_Crossbar.kicad_sch

Title: Passive module | 140 contacts

Size: A2	Date:	Rev: C-review
KiCad E.D.A. 10.0.8		Id: 123/73